PGCP-AC

INTRODUCTION TO APPLIED GENERATIVE AI: FROM LLMS TO AGENTIC AI

Foundations of AI, NLP & Modern LLMs

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What is Artificial Intelligence?

- **Artificial Intelligence (AI) refers to computer systems capable of performing tasks that normally require human intelligence.**

- **Examples:**
  - Image recognition
  - Speech recognition
  - Language understanding
  - Decision making
  - Content generation



Applications of AI-powered DSS

Healthcare:

Finance:

Retail & E-commerce:

Supply Chain Management:

Manufacturing:

Customer Service:

Real-World Applications

Industry	Application	Example
Healthcare	Disease diagnosis	Detecting diabetes or cancer from patient data
Banking	Fraud detection	Identifying suspicious transactions
E-commerce	Product recommendations	Amazon suggesting related products
Education	Personalized learning	Recommending learning resources based on student performance
Transportation	Traffic prediction	Google Maps estimating travel time
Agriculture	Crop prediction	Predicting crop yield and detecting plant diseases
Manufacturing	Quality inspection	Detecting defective products on production lines
Entertainment	Content recommendation	Netflix and Spotify recommendations

Input X

Output y class

Not hearing anything? Turn up volume ×

Age	Salary	Purchased
25	30K	No
40	70K	Yes
35	50K	Yes

Predictor

Ask : "Predict whether a person will buy a produ

You are screen sharing Stop share

Machine Learning

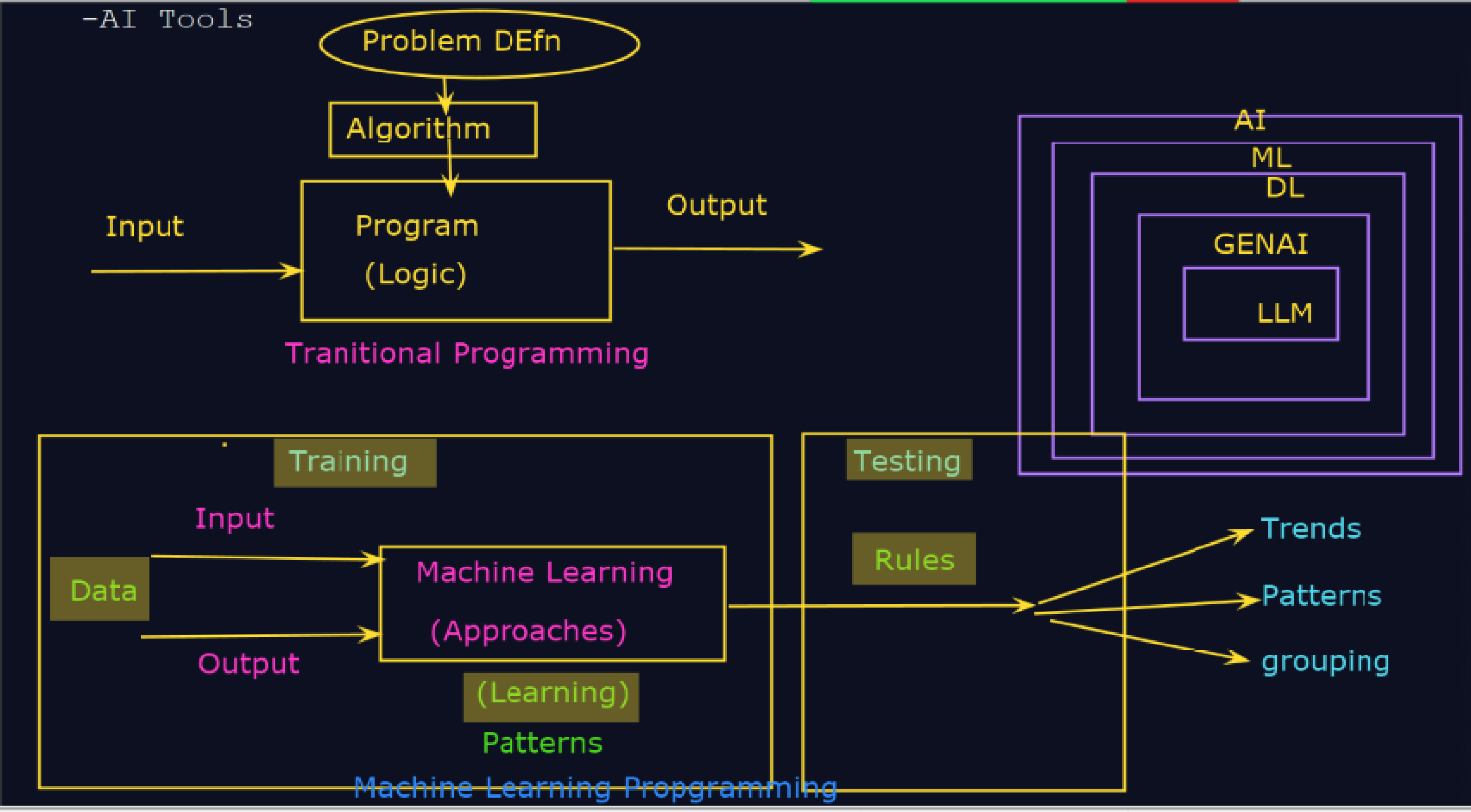


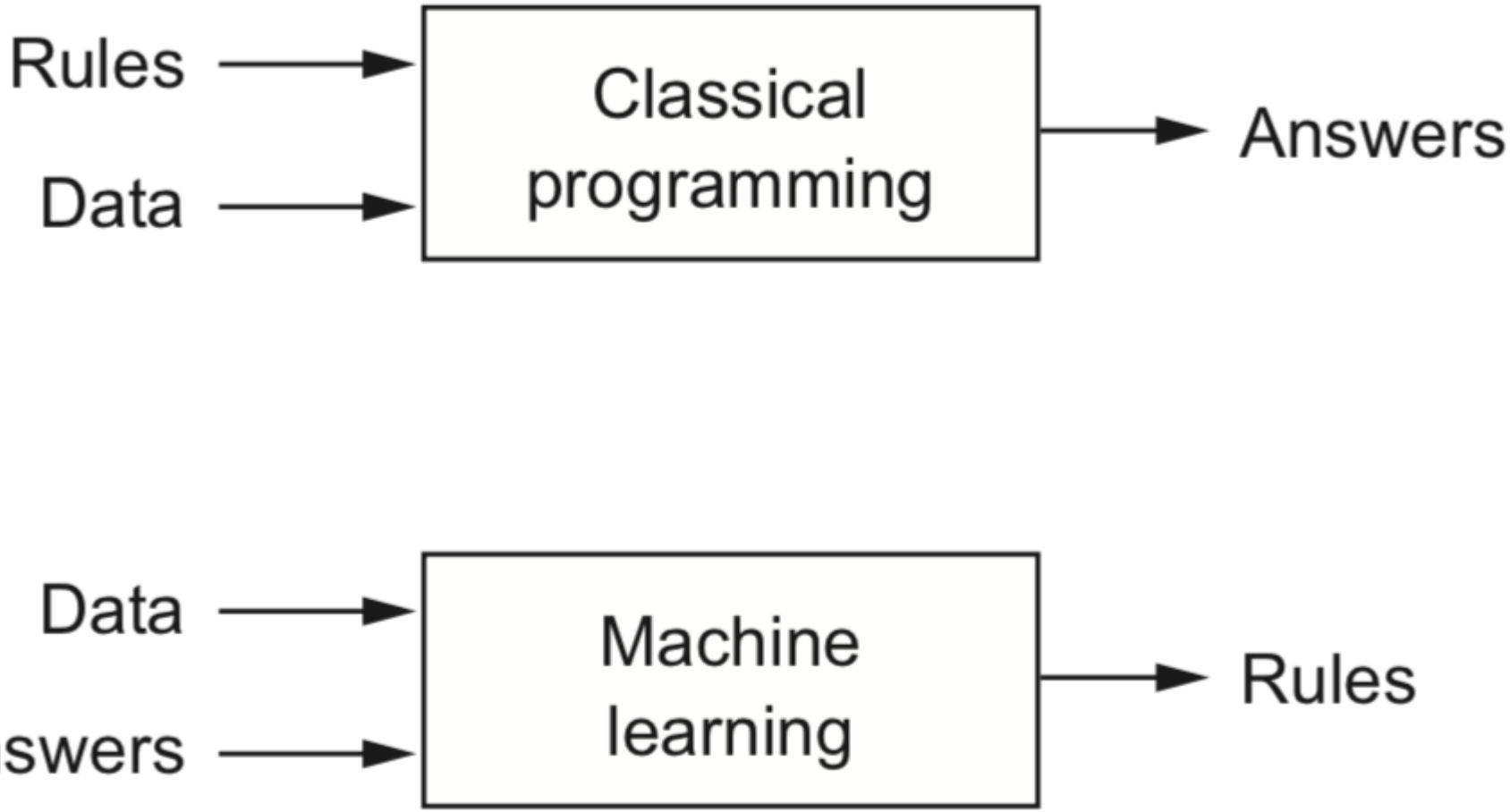


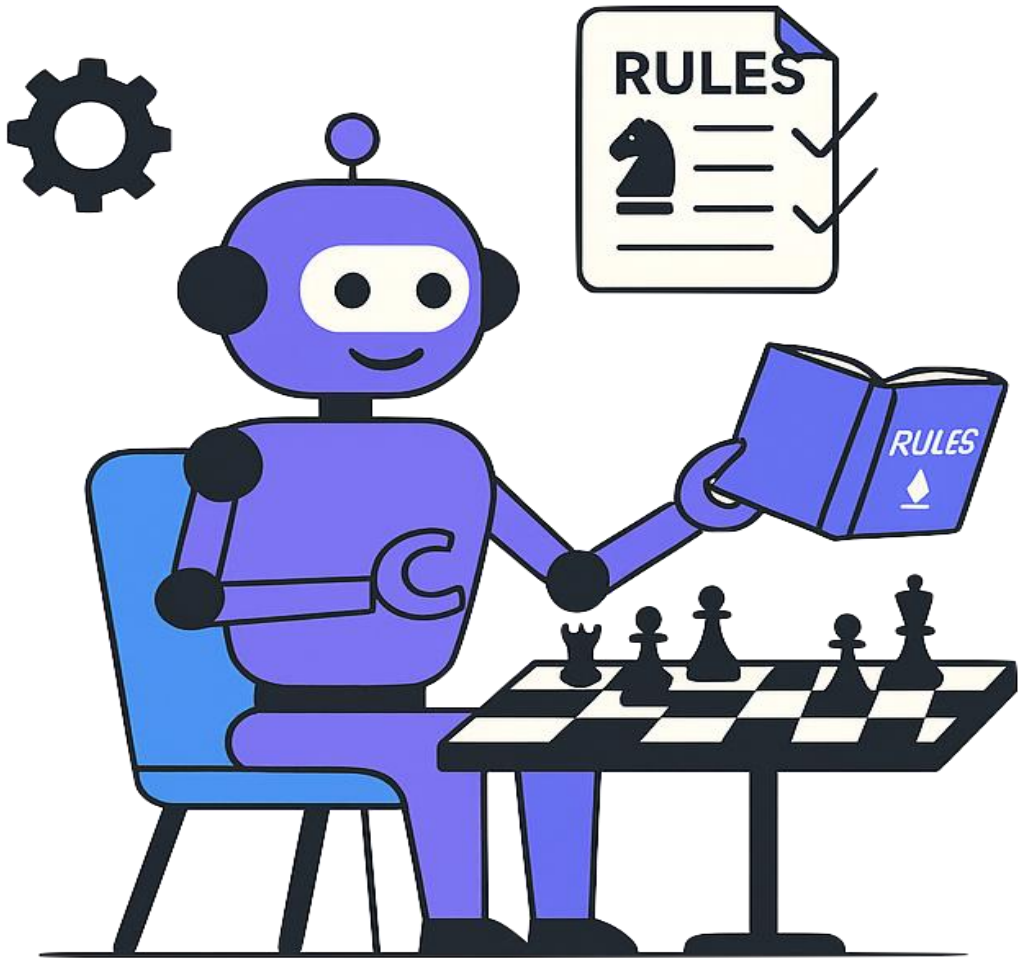






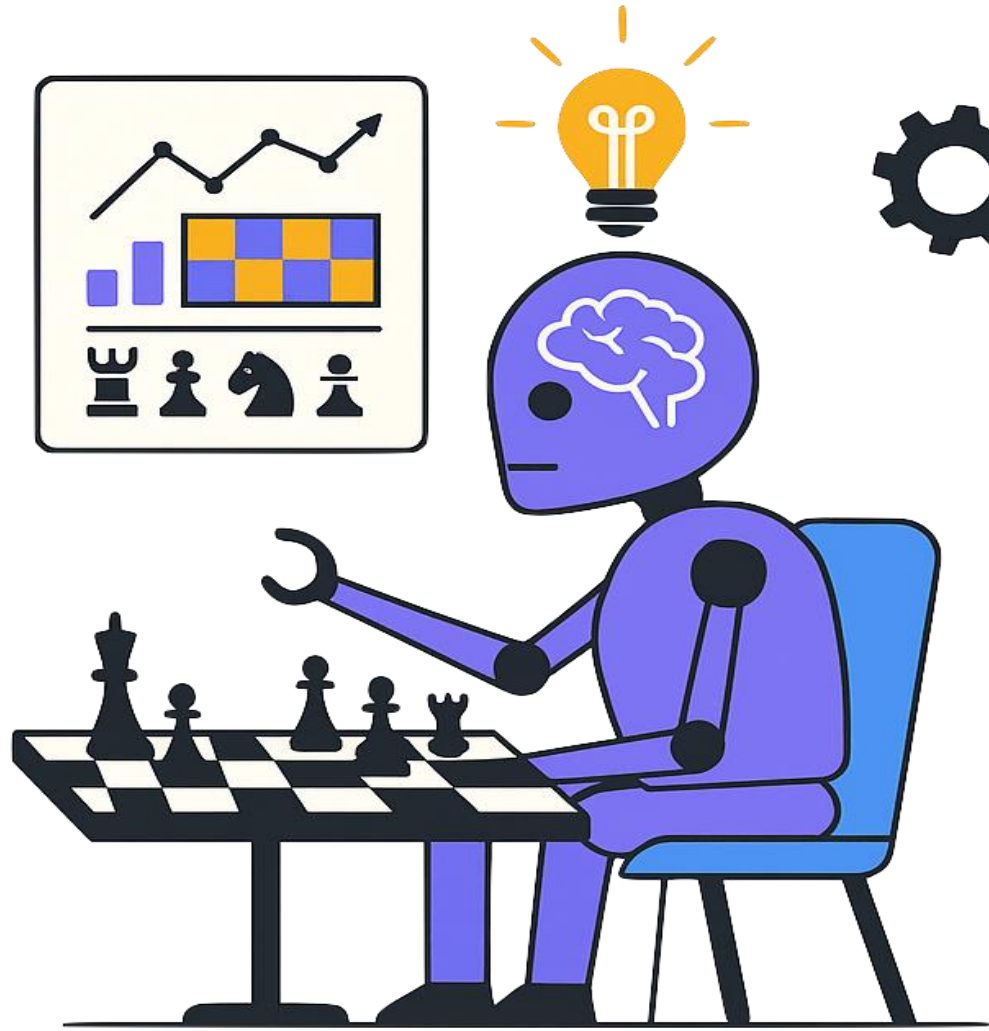

-AI Tools
Problem DEfn
Algorithm
Input
Program
(Logic)
Output
Tranitional Programming
AI
ML
DL
GENAI
LLM
Training
Input
Data
Machine Learning
(Approaches)
Output
(Learning)
Patterns
Machine Learning Propgramming
Testing
Rules
Trends
Patterns
grouping


Rules
Data
Classical programming
Answers
Data
Answers
Machine learning
Rules



Traditional AI

follows programmed strategies

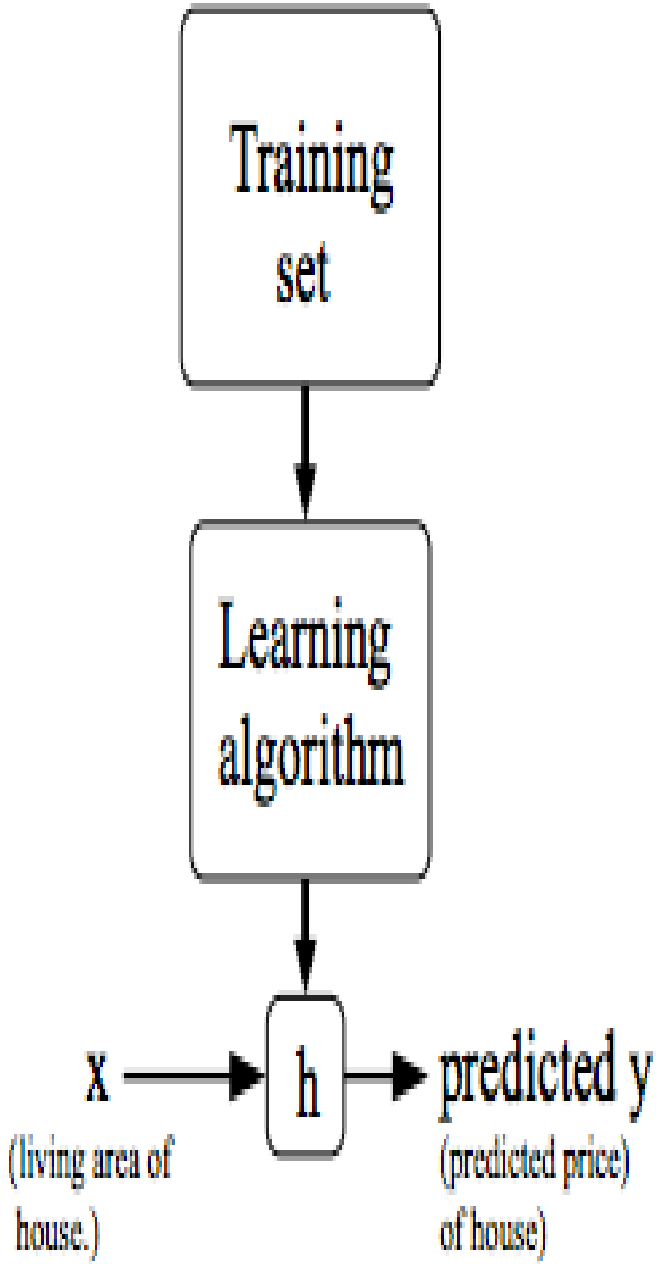


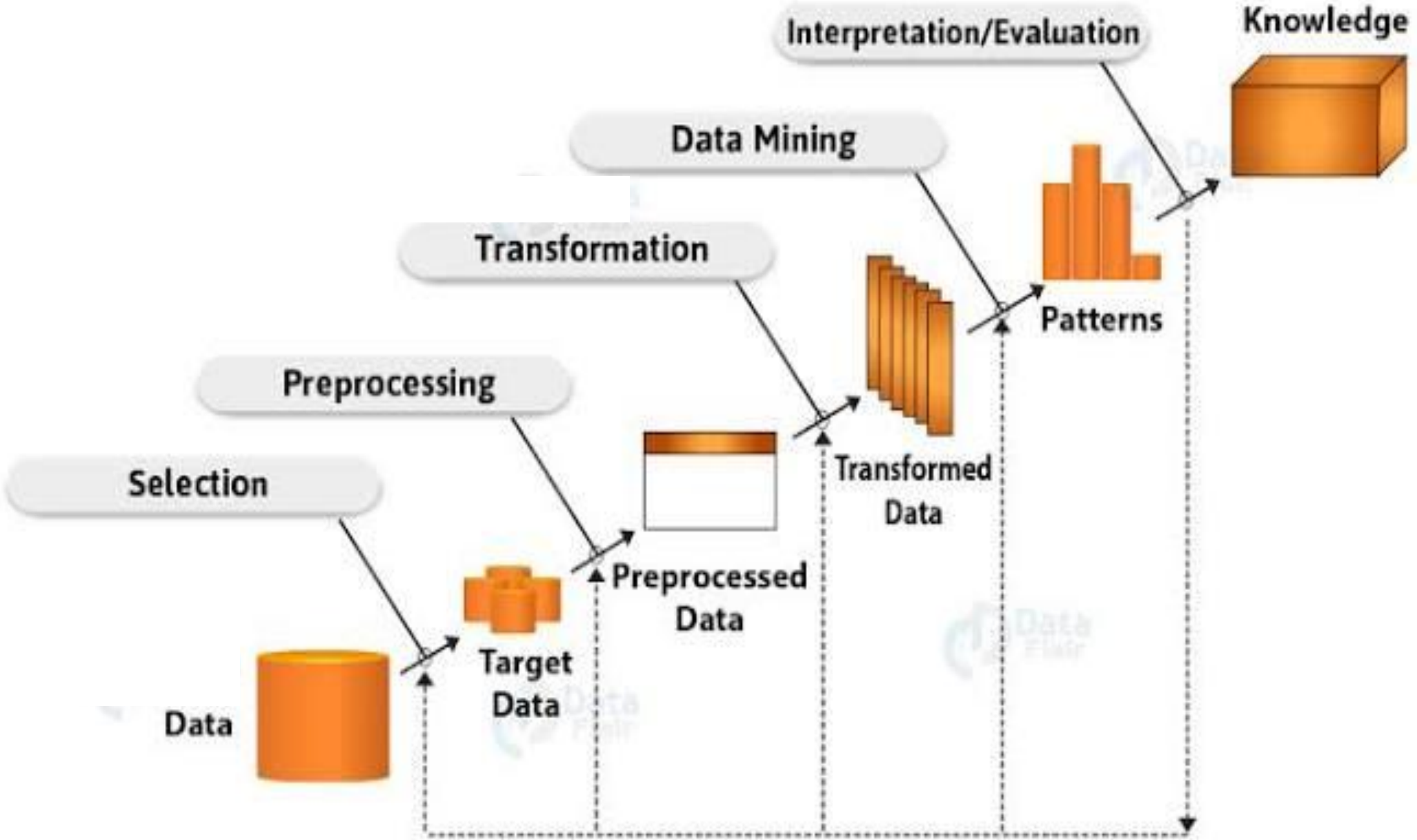
Machine Learning

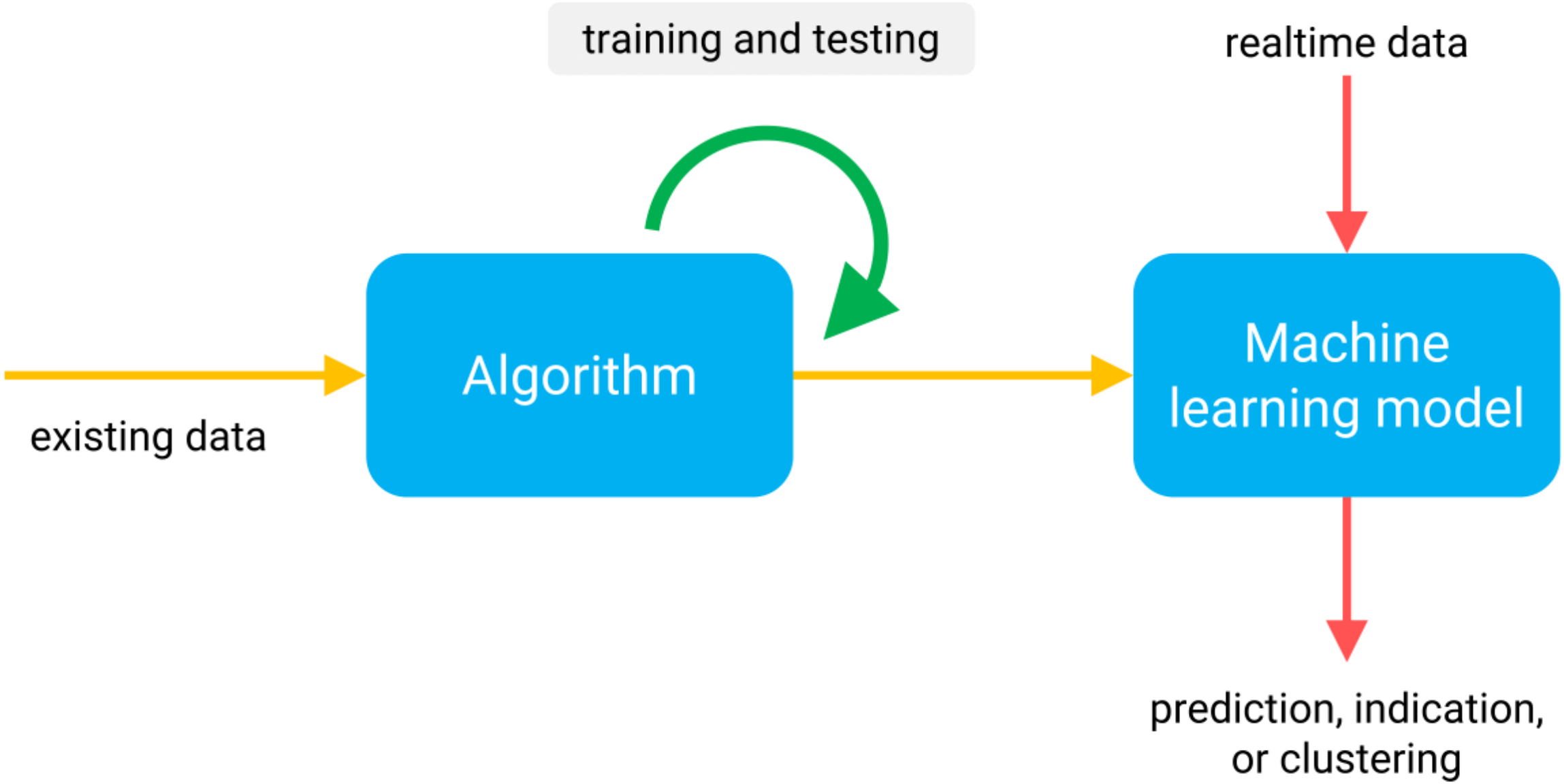
learns from data and experience

Machine Learning

- **Machine Learning (ML)Machine Learning (ML) is a branch of Artificial Intelligence (AI)** that enables computers to learn from data and improve their performance without being explicitly programmed.
- Instead of following fixed rules, machine learning algorithms identify patterns in data and use those patterns to make predictions or decisions.
- **Definition**
  - Machine Learning is the process of training computers to learn from historical data and make predictions or decisions automatically.

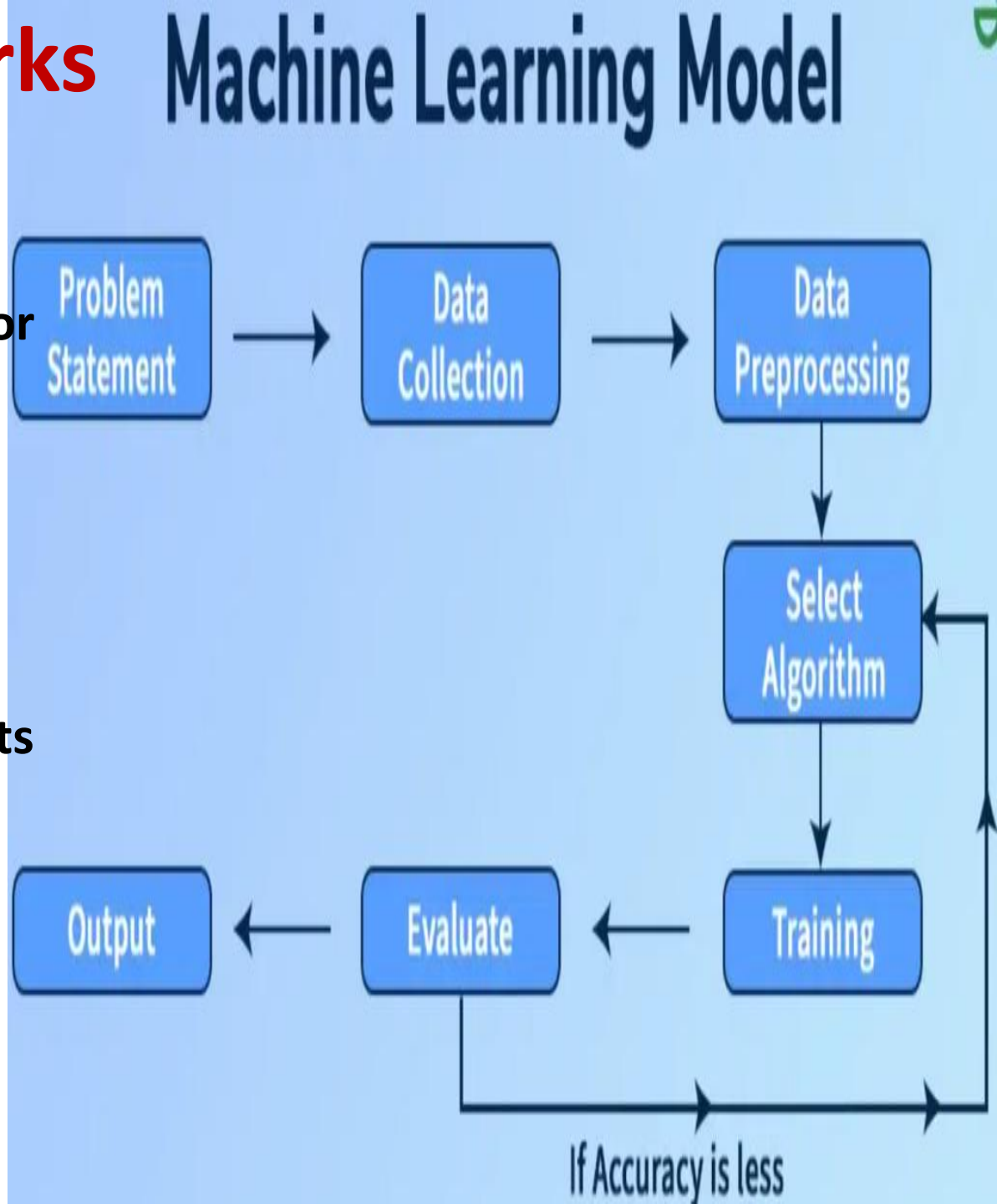



Interpretation/Evaluation
Knowledge
Data Mining
Transformation
Patterns
Preprocessing
Transformed Data
Selection
Preprocessed Data
Target Data
Data


training and testing
realtime data
Algorithm
existing data
Machine learning model
prediction, indication, or clustering

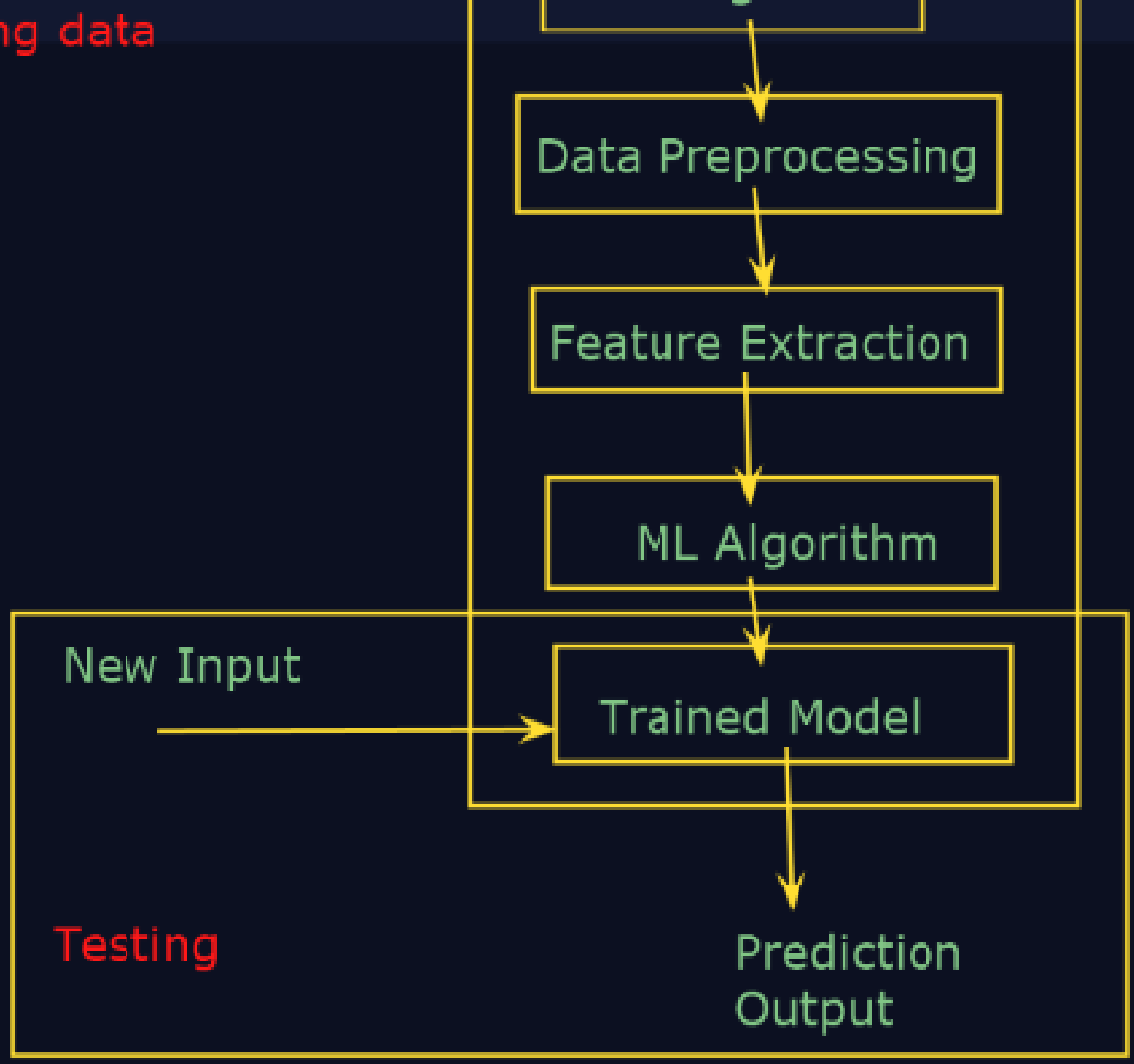
How Machine Learning Works

1. Collect Data – Gather relevant data (text, images, numbers, etc.).
2. Prepare Data – Clean and organize the data for training.
3. Train the Model – Feed the data into a machine learning algorithm.
4. Learn Patterns – The algorithm identifies relationships and patterns in the data.
5. Make Predictions – The trained model predicts outcomes for new, unseen data.
6. Evaluate and Improve – Test the model's accuracy and retrain if necessary.



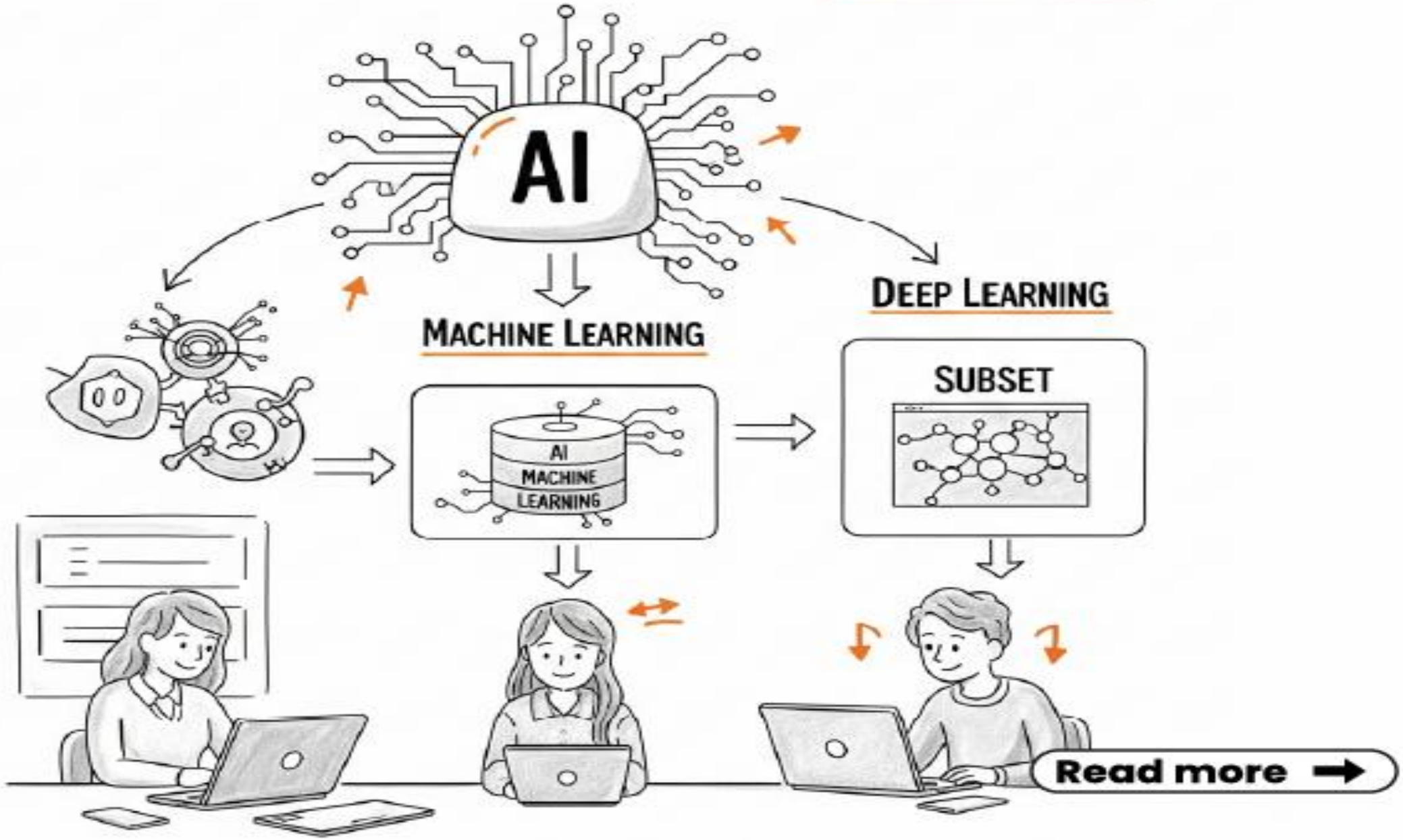
ML: It enables computers to learn from data and improve their performance being explicitly programmed.

Working:

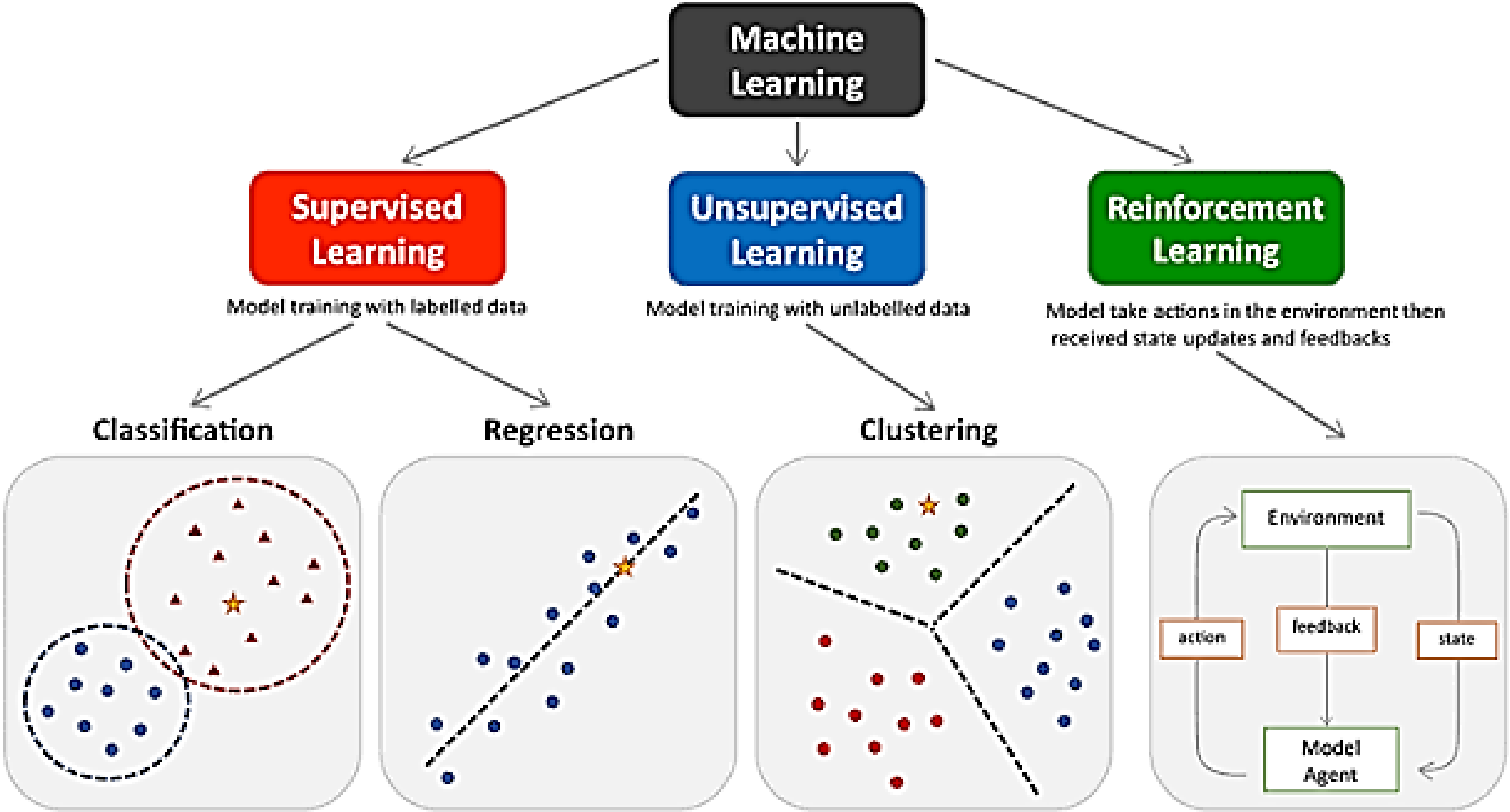


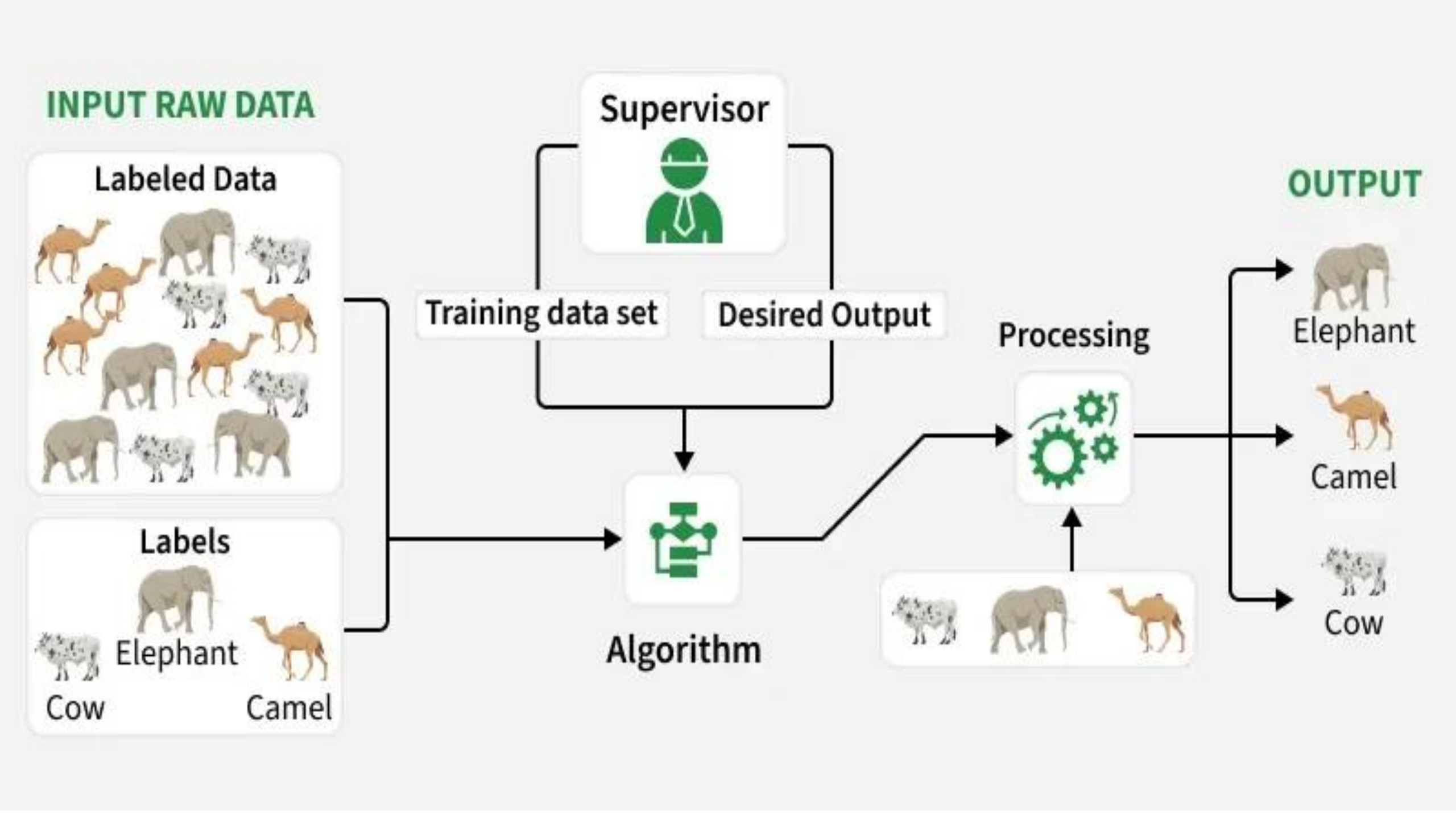
AI vs. Machine Learning vs. Deep Learning:

Explained Simply



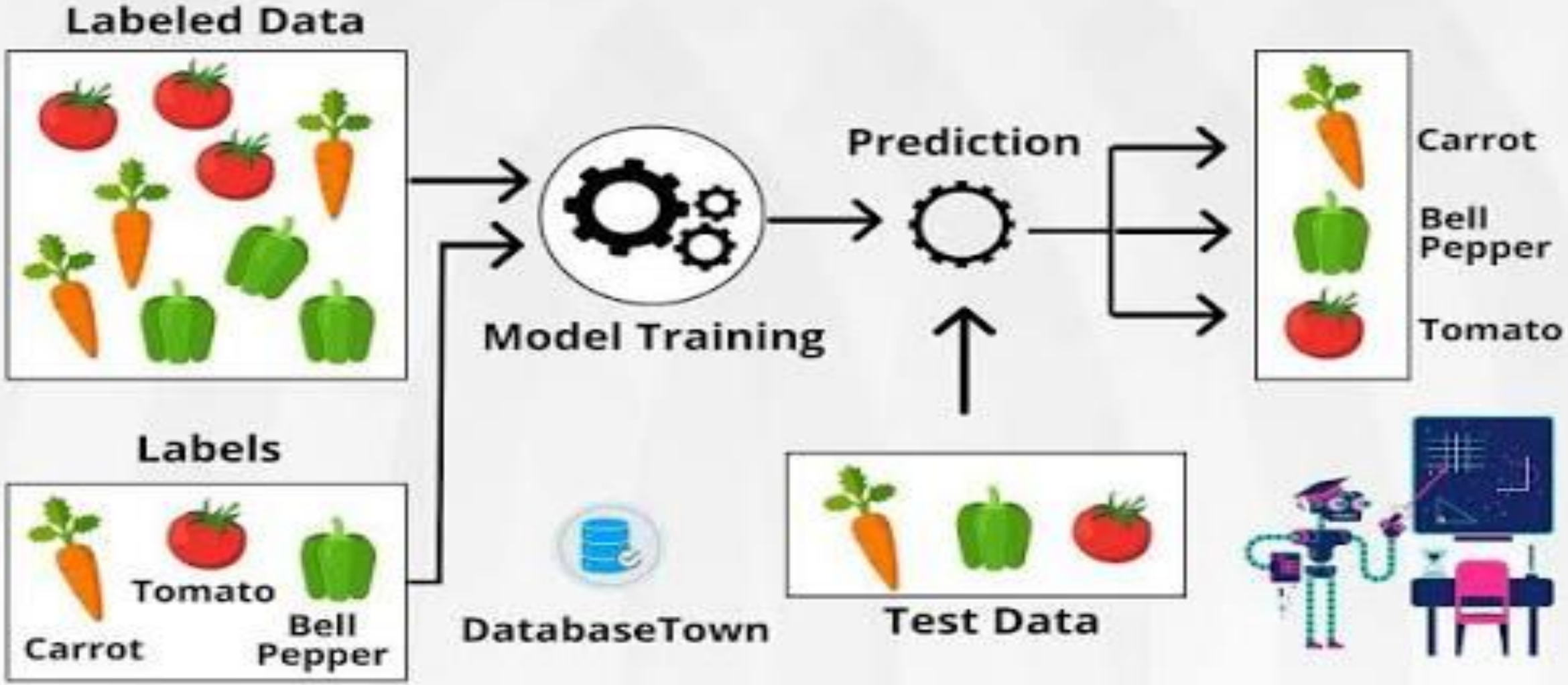

EMAIL FILTER
Inbox
Spam


Machine Learning
Supervised Learning
Unsupervised Learning
Reinforcement Learning
Model training with labelled data
Model training with unlabelled data
Model take actions in the environment then received state updates and feedbacks
Classification
Regression
Clustering
Environment
action
feedback
state
Model
Agent


INPUT RAW DATA
Labeled Data
Labels
Elephant
Cow
Camel
Supervisor
Training data set
Desired Output
Algorithm
Processing
OUTPUT
Elephant
Camel
Cow

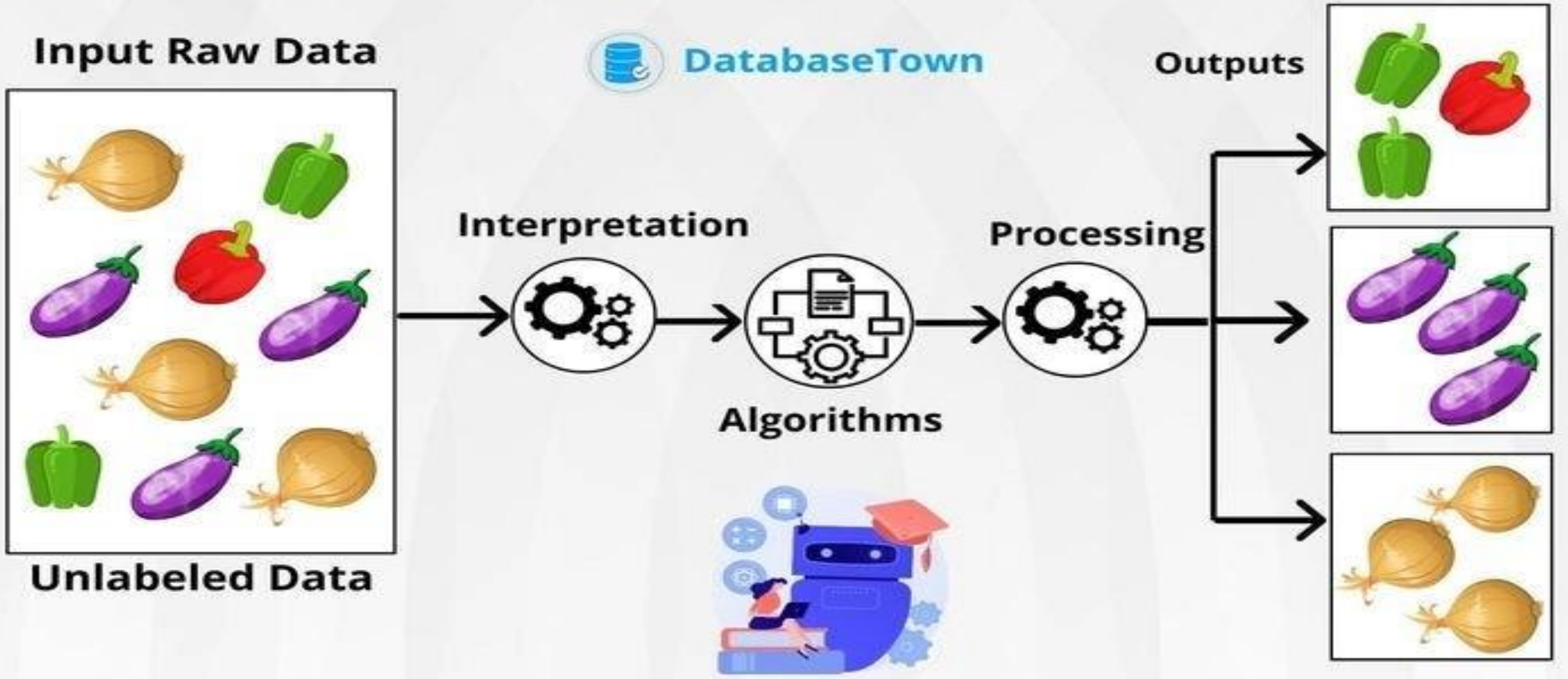
SUPERVISED LEARNING

Supervised machine learning is a branch of artificial intelligence that focuses on training models to make predictions or decisions based on labeled training data.



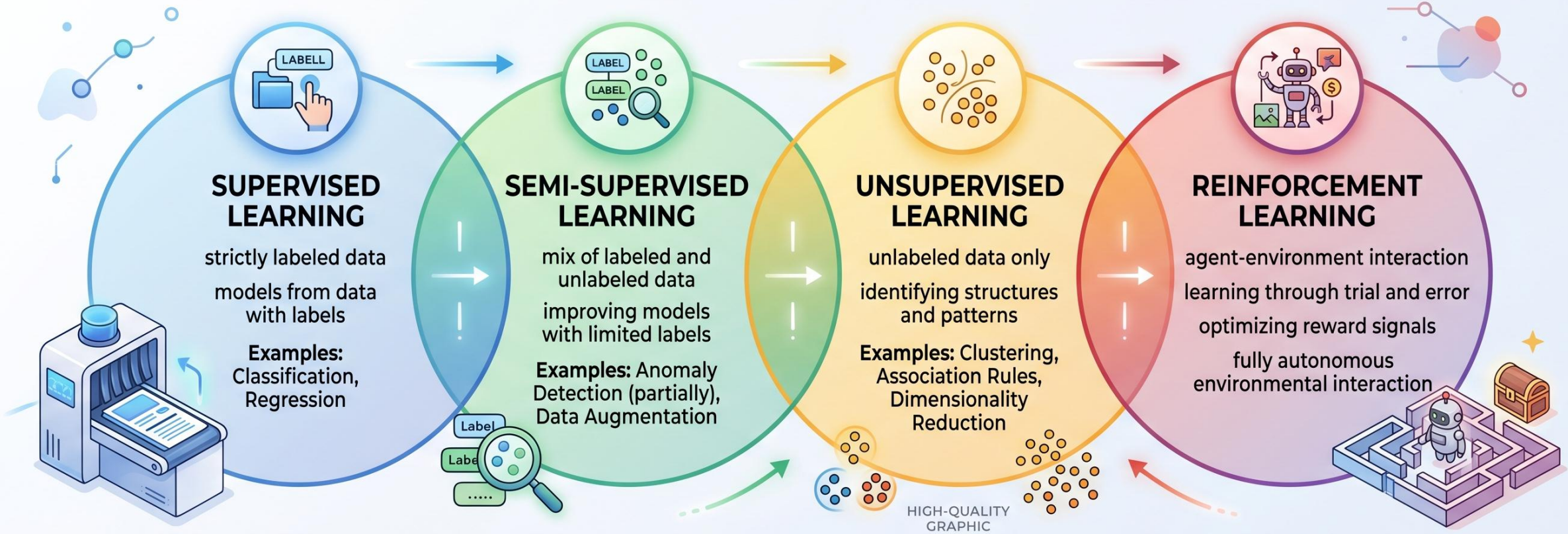
UNSUPERVISED LEARNING

Unsupervised learning is a type of machine learning where the algorithm learns from unlabeled data without any predefined outputs or target variables.



A SPECTRUM OF MACHINE LEARNING PARADIGMS

FROM LABELED DATA TO AUTONOMOUS INTERACTION



INCREASING AUTONOMY & INTERACTION

DECREASING DEPENDENCE ON LABELED DATA

Not hearing anything? Turn up volume ×

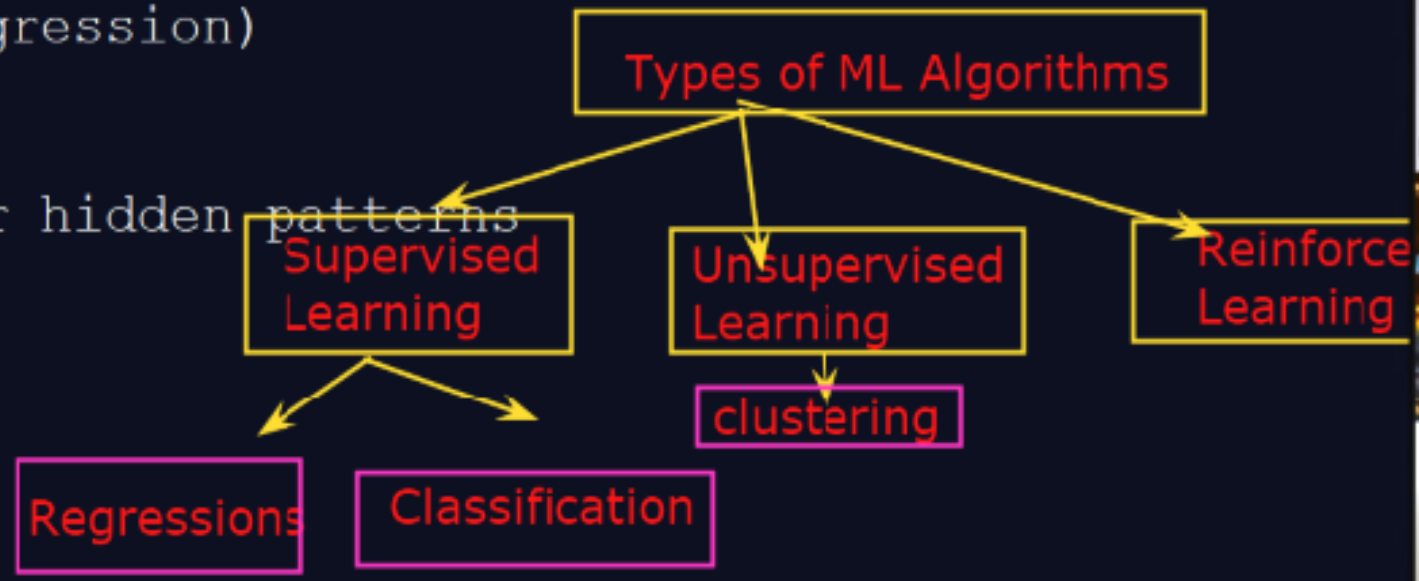
-Ex:
    -Email Spam detection (Classification)
    -House price predictions (Regression)

2.Unsupervised: No labelled data
-Uses unlabelled data to discover hidden patterns

-Ex:
    -customer segmentation
    -Student grouping
    -Market Basket Analysis

3.Reinforcement: trial and error
-Learns through rewards and penalties by interacting with an environment
-Ex:
    -Self driving cars
    -Robotics
    -Game playing AI

4.semi supervised: supervised+ unsupervised



You are screen sharing
Stop share

Uses of Machine Learning Models

Machine Learning (ML) models learn patterns from data and use those patterns to make predictions, classify information, detect anomalies, or recommend actions. They are widely used across industries to automate decision-making and improve accuracy.

Use of ML Model	Description	Example / Use Case
Classification	Classifies data into predefined categories.	Email **Spam** or **Not Spam**
Prediction	Predicts future values based on historical data.	House price prediction, weather forecasting
Recommendation Systems	Suggests products, movies, or content based on user preferences.	Netflix movie recommendations, Amazon product suggestions
Fraud Detection	Detects unusual or suspicious activities.	Credit card fraud detection in banks
Image Recognition	Identifies objects, people, or scenes in images.	Face recognition in smartphones
Speech Recognition	Converts spoken language into text.	Google Voice Typing, Siri, Alexa
Sentiment Analysis	Determines whether text is positive, negative, or neutral.	Customer review analysis
Medical Diagnosis	Assists doctors by identifying diseases from medical data.	Detecting tumors in X-ray or MRI images
Customer Segmentation	Groups customers with similar characteristics.	Targeted marketing campaigns
Predictive Maintenance	Predicts when equipment is likely to fail.	Monitoring machines in factories to schedule maintenance

Common Types of Machine Learning Models

Model Type	Purpose	Example
Linear Regression	Predict continuous values	Predicting house prices
Logistic Regression	Binary classification	Spam vs. Not Spam
Decision Tree	Classification and prediction	Loan approval
Random Forest	Improves prediction accuracy using multiple decision trees	Fraud detection
Support Vector Machine (SVM)	Classifies complex datasets	Image classification
K-Nearest Neighbors (KNN)	Classifies based on similar data points	Handwritten digit recognition
Naïve Bayes	Text classification	Spam email detection
Neural Networks	Learns complex patterns	Speech and image recognition

Deep Learning

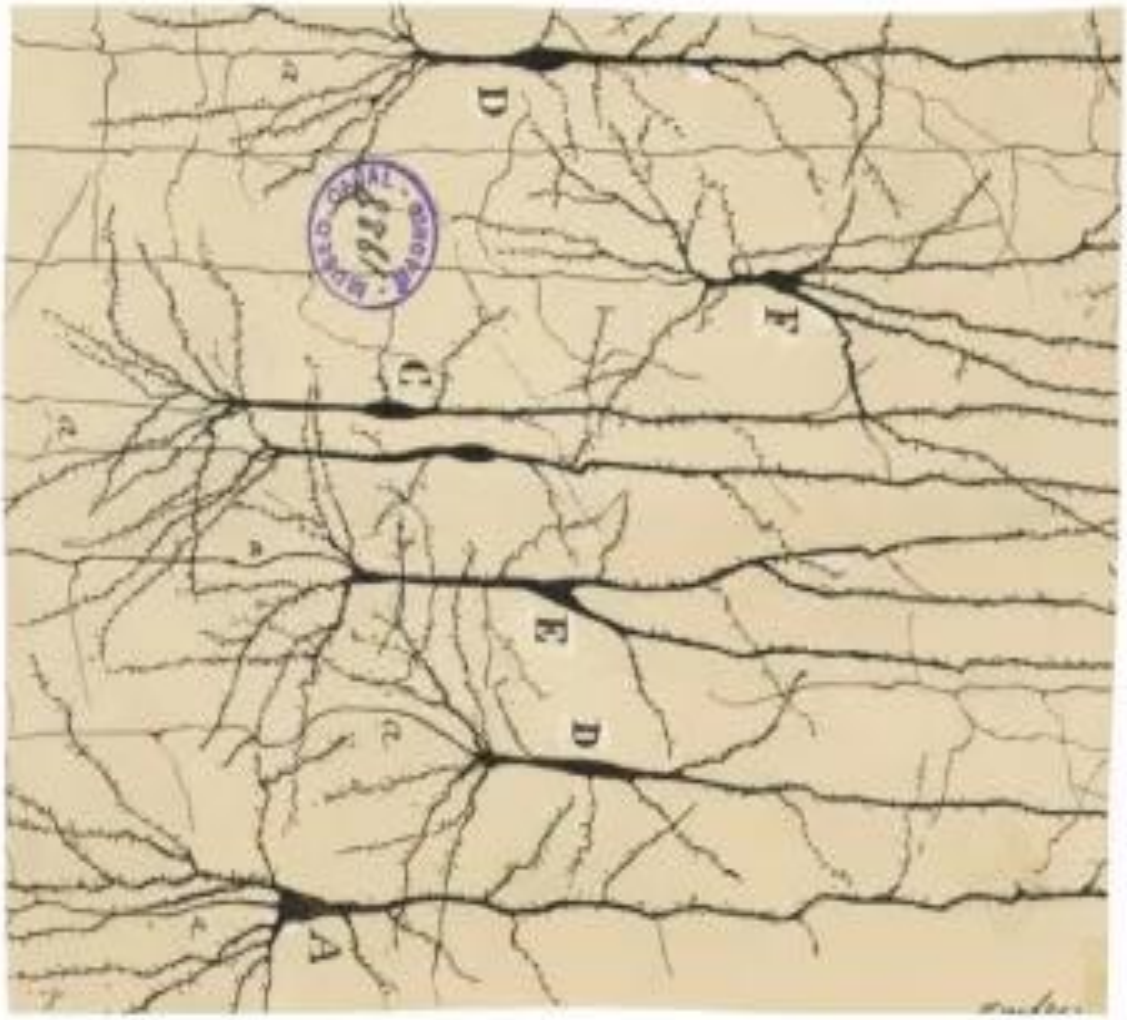


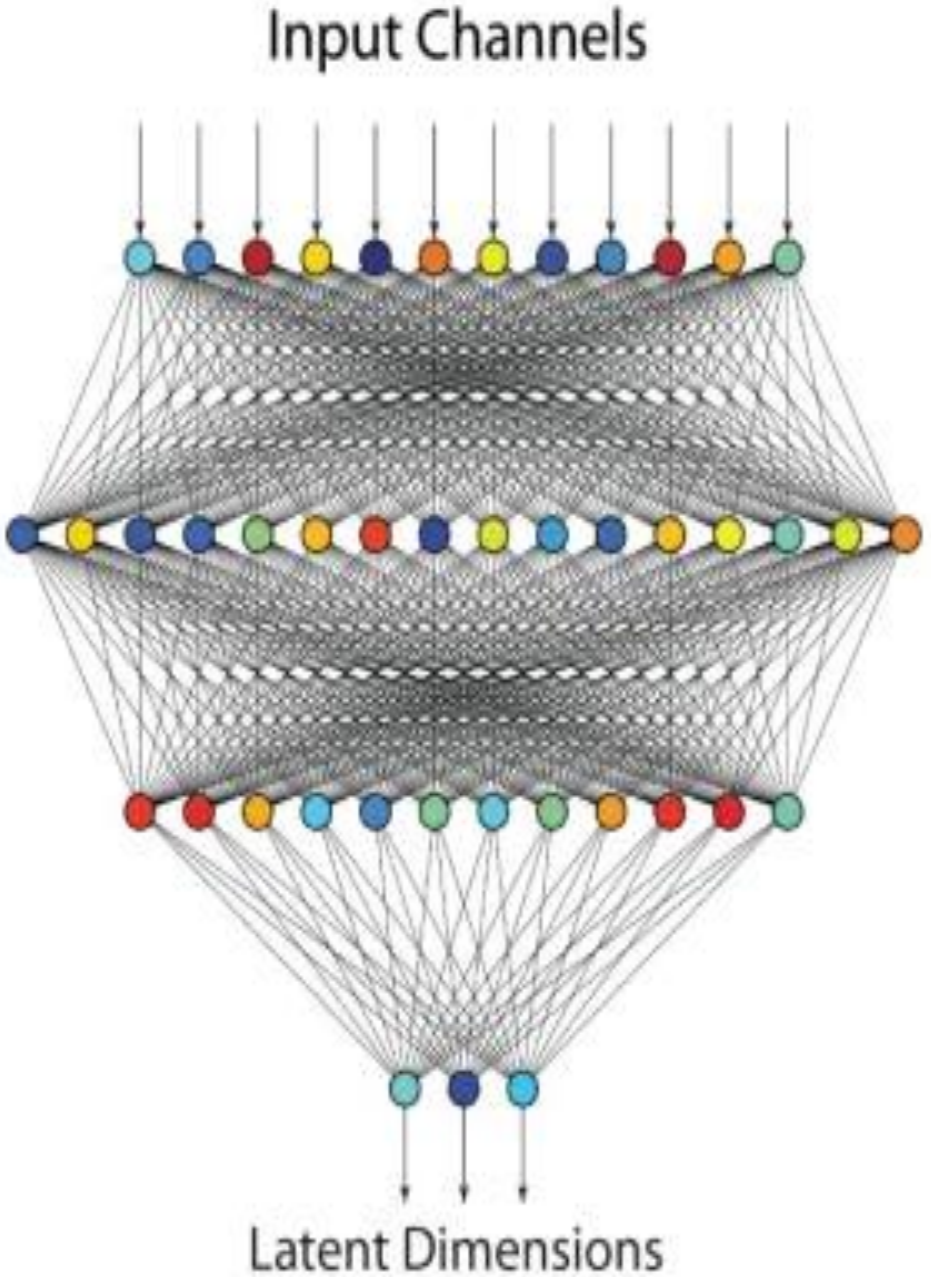


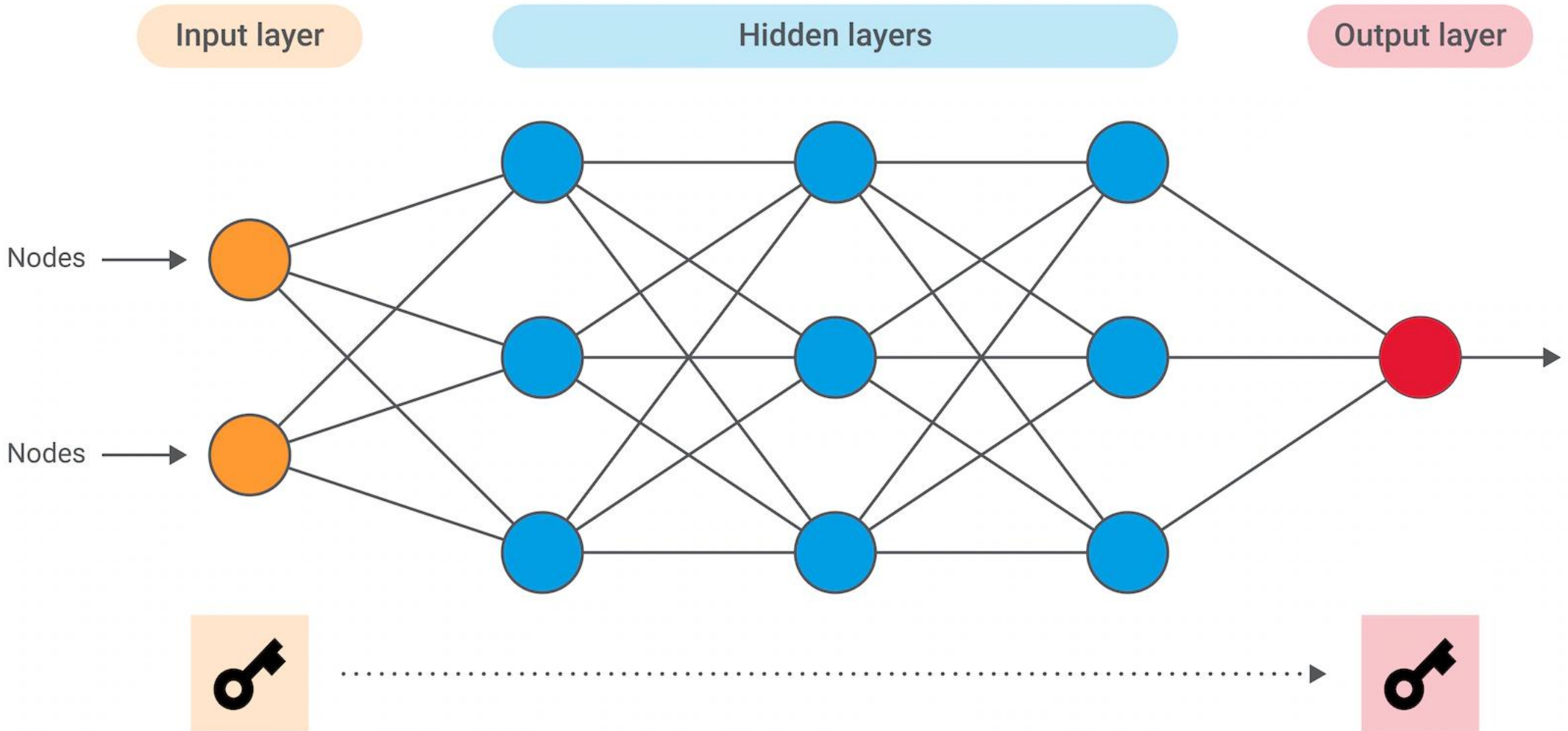








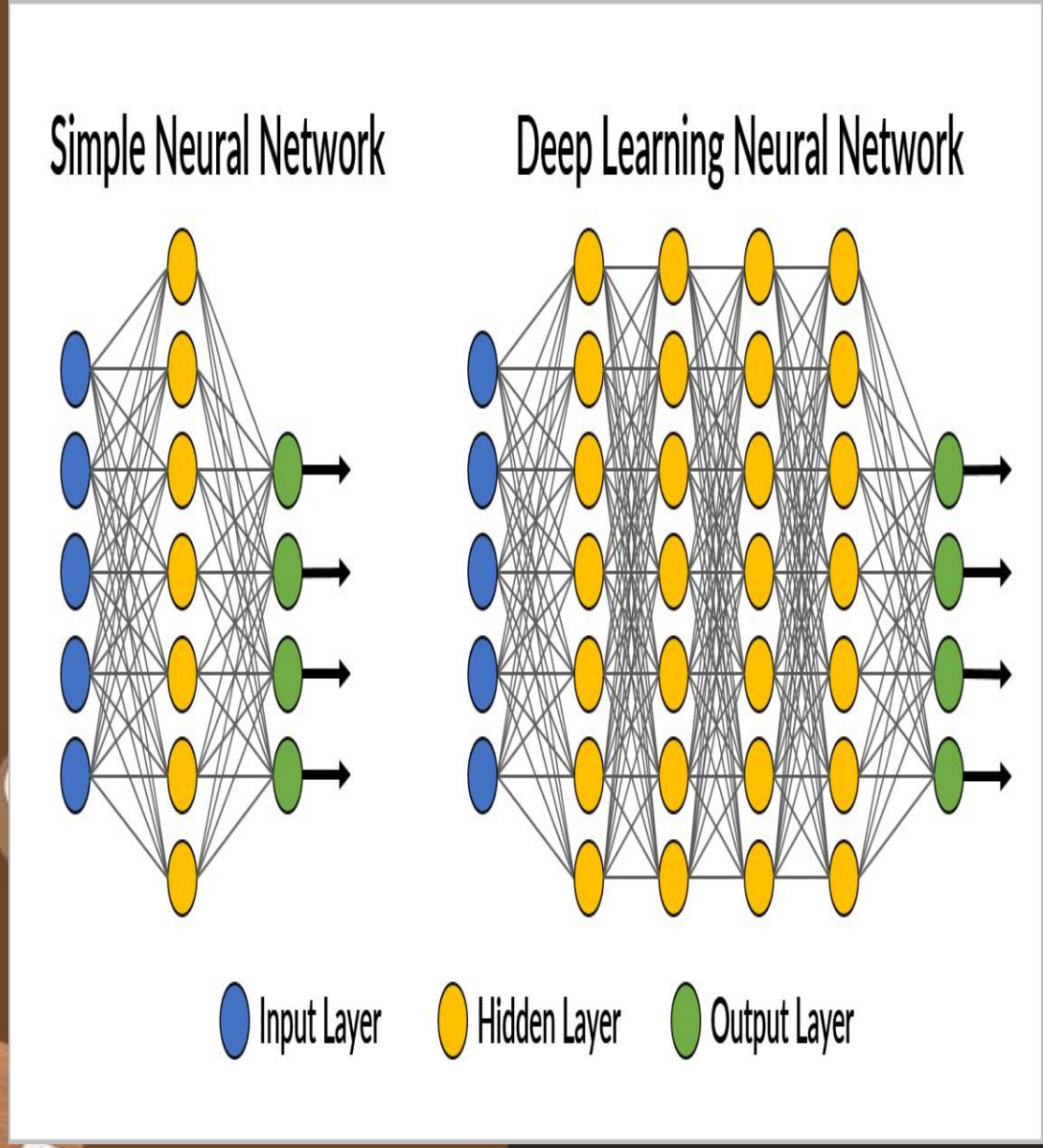

Input Channels
Latent Dimensions


Input layer
Hidden layers
Output layer
Nodes
Nodes
"What is this image of?"
"This is an image of a key"

Deep Learning

- **Deep Learning (DL) is a subset of Machine Learning (ML) that uses Artificial Neural Networks (ANNs) with multiple layers to learn complex patterns from large amounts of data.**
- **It is inspired by the way the human brain processes information through interconnected neurons.**

- **Definition**
  - Deep Learning is a branch of Machine Learning that uses multi-layer neural networks to automatically learn features and make predictions from large datasets.








How to Design a Neural Network

Neural networks pass an input, like an image, through multiple layers of digital neurons. Each layer reveals additional features of the input. Mathematicians are revealing how a network's architecture — how many neurons and layers it has and how they're connected — determines the kinds of tasks that the neural network will be good at.



When data is fed into a network, each artificial neuron that fires (labeled "1") transmits signals to certain neurons in the next layer, which are likely to fire if multiple signals are received. This process reveals abstract information about the input.



A SHALLOW NETWORK has few layers but many neurons per layer. These "expressive" networks are computationally intensive.

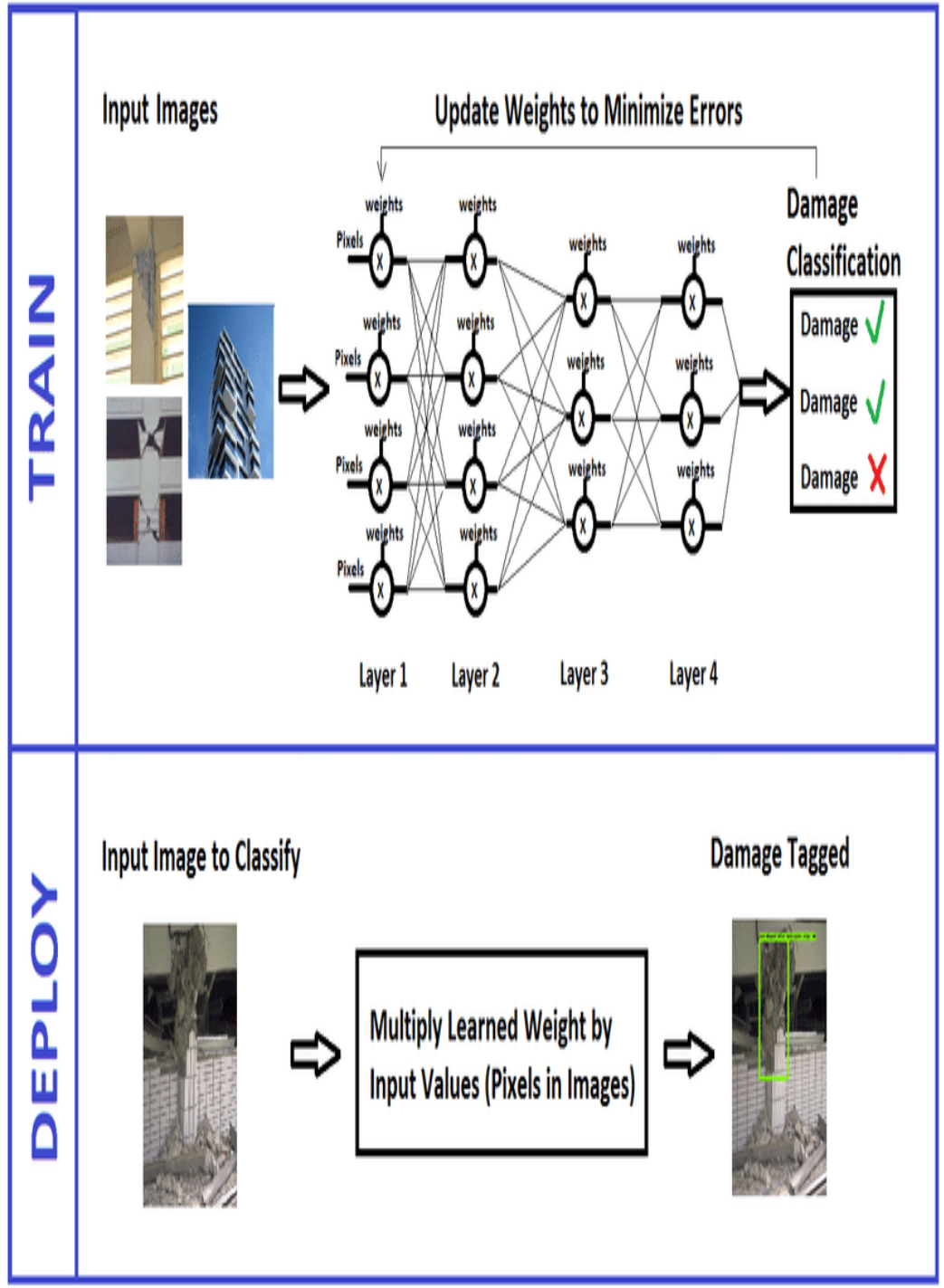


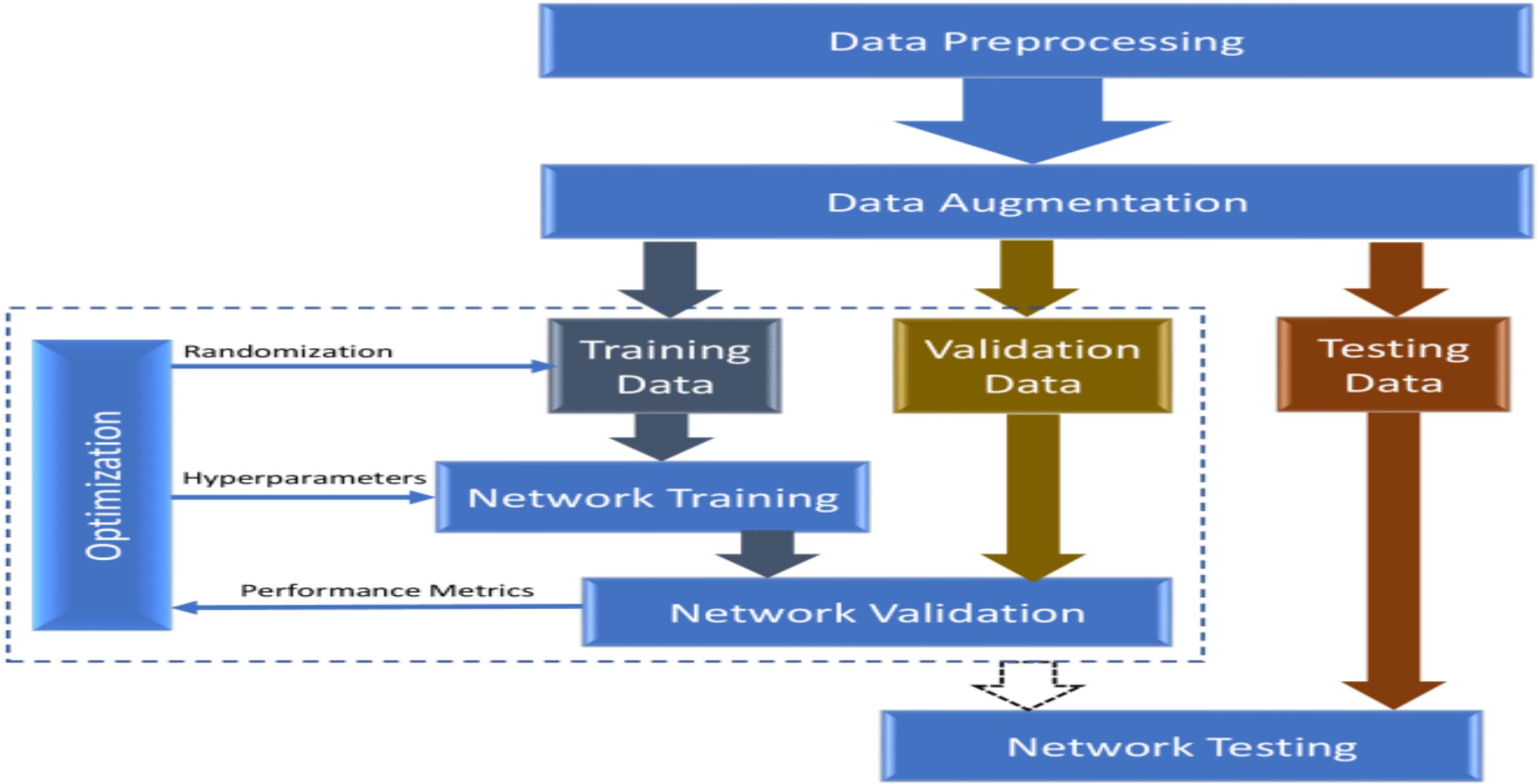
A DEEP NETWORK has many layers and relatively few neurons per layer. It can achieve high levels of abstraction using relatively few neurons.

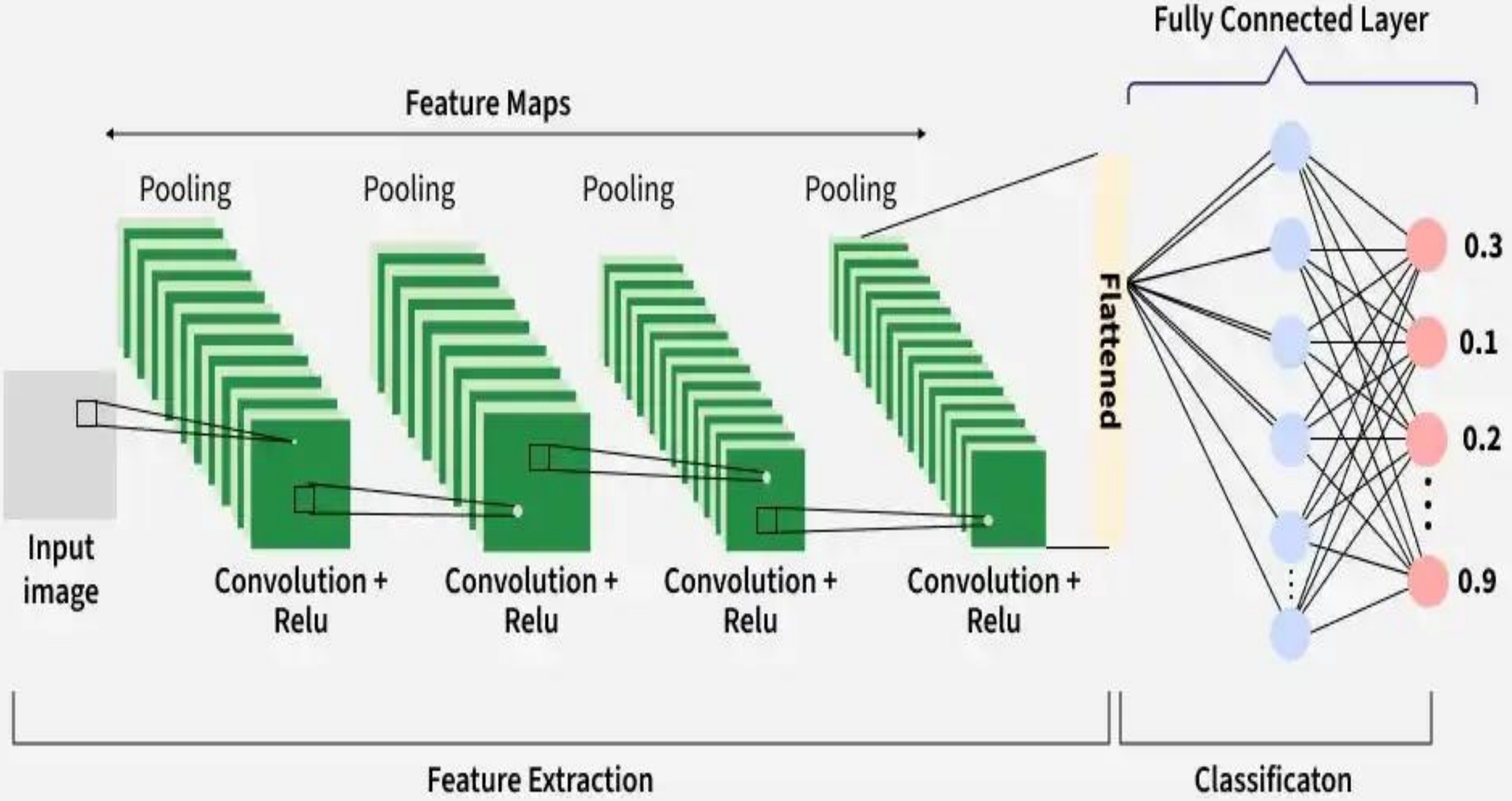
How Deep Learning Works

- **Collect Data**
  - Gather large amounts of data (images, text, audio, videos, etc.).
- **Input Layer**
  - Data is fed into the neural network.
- **Hidden Layers**
  - Multiple hidden layers process the data and learn increasingly complex patterns.
- **Output Layer**
  - Produces the final prediction or classification.
- **Training**
  - The model compares its predictions with the correct answers.
  - Errors are reduced using **backpropagation** and **optimization algorithms**.
- **Prediction**
  - Once trained, the model can predict outcomes for new, unseen data.






Data Preprocessing
Data Augmentation
Training Data
Validation Data
Testing Data
Optimization
Randomization
Hyperparameters
Performance Metrics
Network Training
Network Validation
Network Testing


Fully Connected Layer
Feature Maps
Pooling
Pooling
Pooling
Pooling
Flattened
0.3
0.1
0.2
0.9
Input image
Convolution + Relu
Convolution + Relu
Convolution + Relu
Convolution + Relu
Feature Extraction
Classificaton

Machine Learning vs Deep Learning

Feature	Machine Learning	Deep Learning
Subset	AI	Machine Learning
Feature Engineering	Mostly manual	Automatic
Data Requirement	Moderate	Very High
Accuracy	Good	Excellent on complex tasks
Training Time	Shorter	Longer
Hardware	CPU often sufficient	GPUs/TPUs preferred
Interpretability	Higher	Lower
Best For	Structured data	Images, audio, video, text

Generative AI

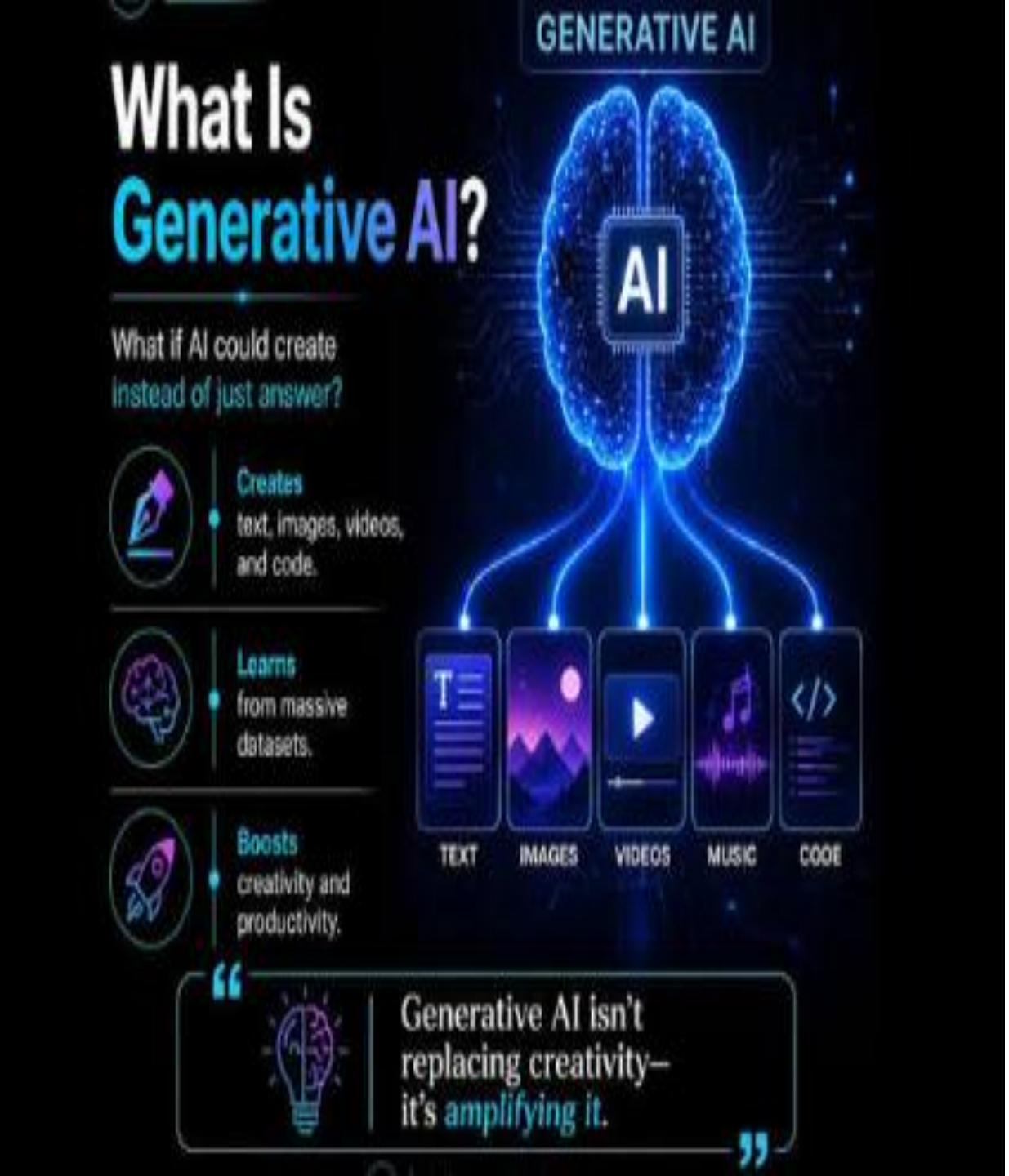












Generative AI (Gen AI)

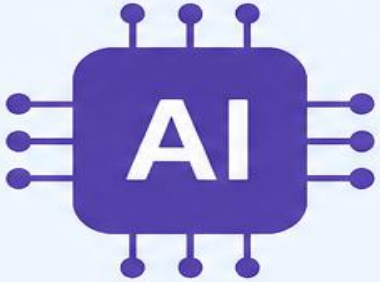
- Generative AI is a type of Artificial Intelligence that **creates new and original content instead of only analyzing or classifying existing data.**
- It **generates content based on a prompt** (text, image, audio, video, or code) provided by the user.
- Unlike Traditional AI, which mainly predicts or classifies, Generative AI produces text, images, music, videos, computer code, and other creative outputs.

Generative AI Example

Gen AI creates new content based on your input (prompt).

Input (Prompt)

Write an email requesting leave.



Generative AI

Processes the prompt
and generates content.

Output (Generated Email)

To: manager@example.com

Subject: Request for Leave

Dear Manager,

I am writing to formally request leave from June 10, 2024, to June 14, 2024, as I will be traveling out of town for a family event.

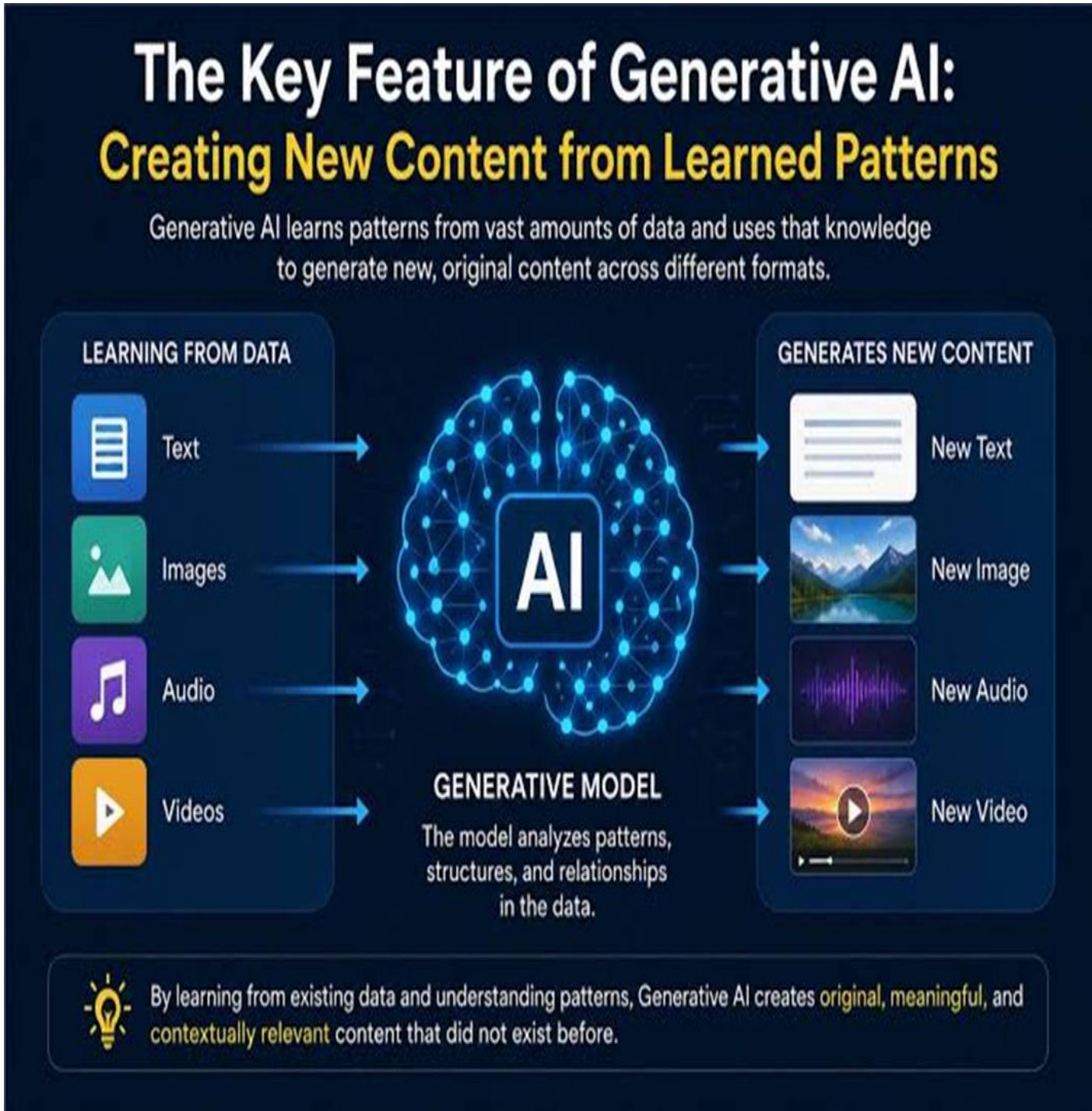
I will ensure that all my tasks are completed before my leave and will be available via email in case of any urgent matters.

Thank you for considering my request.

Sincerely,
Riya Sharma
Marketing Executive

Key Features

- Creates new content rather than simply identifying patterns.
- Understands natural language prompts.
- Produces human-like text and conversations.
- Can generate images, videos, audio, and animations.
- Writes programming code in multiple languages.
- Learns from large datasets to generate meaningful outputs.
- Supports multiple languages.
- Improves productivity by automating creative and repetitive tasks.



Types of Content Generated

- Text (emails, articles, reports, summaries)
- Images (logos, posters, artwork)
- Videos (animations, short clips)
- Audio (music, speech, voiceovers)
- Computer code (Python, Java, C++, HTML, etc.)
- Presentations and documents
- Chatbot conversations



How Generative AI Works

From your prompt to intelligent output in 5 simple steps

1

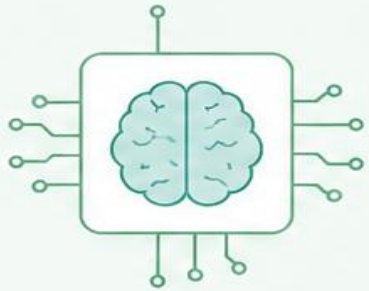
User provides a prompt (instruction or question).



The user enters a prompt in natural language.

2

The AI processes the prompt using a trained AI model.



The model understands the prompt and converts it into a form it can work with.

3

It analyzes patterns learned from massive datasets.



The AI looks at the patterns, examples, and relationships in vast amounts of data.

4

It predicts the most appropriate sequence of words, pixels, sounds, or code.



The model predicts the next best elements (step by step) to form a coherent response.

5

It generates a new and relevant response or content.



The AI generates original, relevant, and useful content as the output.

OUTPUT

In Short

Generative AI turns your ideas and questions into meaningful, original content using the power of advanced AI models and vast data.

Generated Content (Example)

Subject: Request for Leave

Dear Manager,

I am writing to request leave for three days from June 10 to June 12, 2024.
I would be grateful if you could approve my request.
Thank you,
Sincerely,
Your Name

Key Point

The quality of the output depends on the clarity of your prompt and the knowledge of the AI model.

NLP:
Natural Language Processing











Natural Language Processin

- **Definition**
- **Natural Language Processing (NLP)** is a branch of Artificial Intelligence that enables computers to process, understand, interpret, and generate human language in both text and speech.







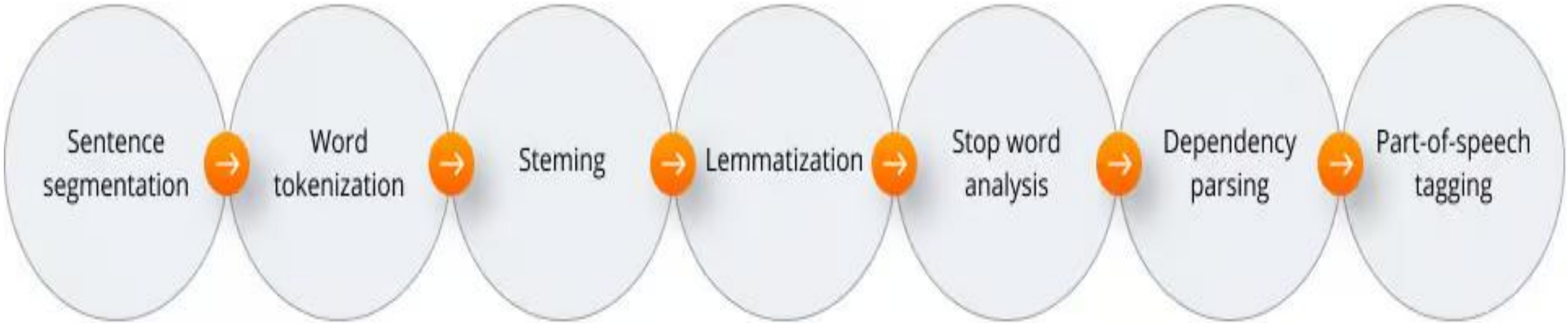

Natural Language Processing (NLP)

- Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that enables computers to understand, interpret, process, and generate human language (text or speech).
- It acts as a bridge between human communication and computer systems, allowing machines to interact with people using natural language instead of programming languages.

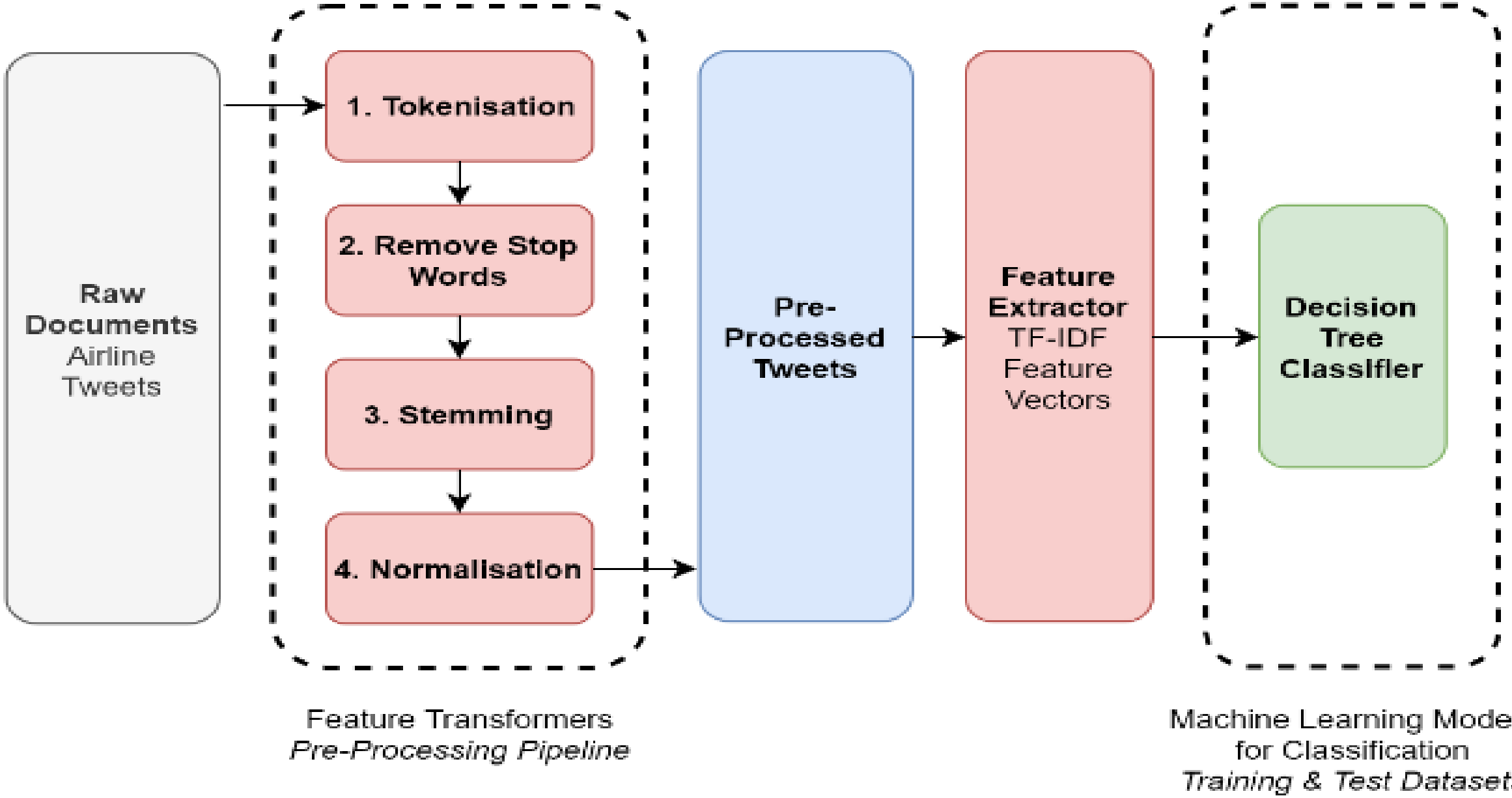
- **Key Objectives of NLP**
- Understand human language (text and speech).
- Analyze the meaning and context of sentences.
- Convert unstructured text into structured information.
- Generate human-like responses.
- Enable natural communication between humans and computers.



Natural Language Processing **Pipeline**



Source: https://www.turing.com/kb/natural-language-processing-function-in-ai


Raw Documents
Airline Tweets
1. Tokenisation
2. Remove Stop Words
3. Stemming
4. Normalisation
Pre-Processed Tweets
Feature Extractor
TF-IDF Feature Vectors
Decision Tree Classifier
Feature Transformers
Pre-Processing Pipeline
Machine Learning Models
for Classification
Training & Test Datasets

Language Models











History of Language Models

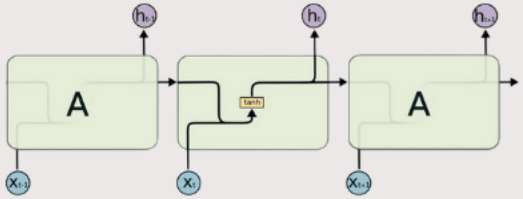
1966

The Dawn of Dialogue — Hello, ELIZA

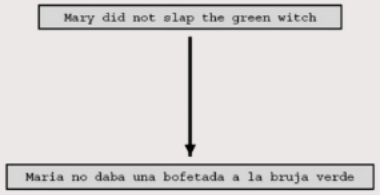


1972

The Rise of Statistical Mechanisms — STNLP and Beyond



Statistical Modelling



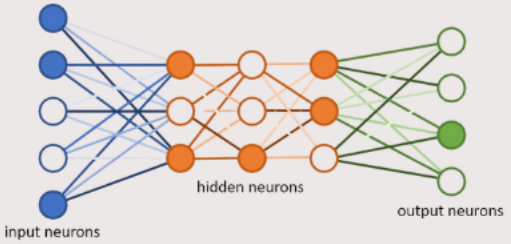
1990s

Memory Revolution — LSTMs

IBM

2000s

An Era of Corporate Innovations



2010s

A Decade of Deep Learning Deluge

2020s

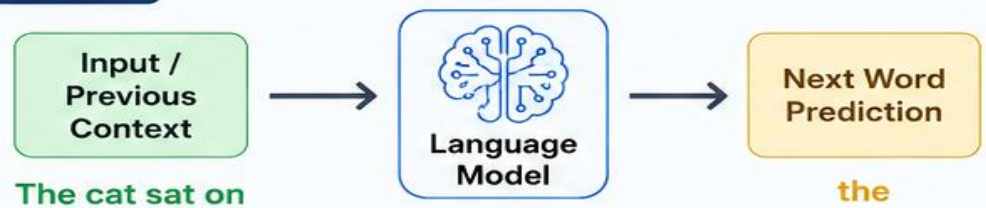
The Era of Giants

OpenAI

LANGUAGE MODEL

A **Language Model (LM)** is an AI model that learns the patterns, structure, and relationships of **human language**. Its primary objective is to **predict the next word (or token) in a sequence** based on the **previous context**.

CONCEPT



The model analyzes the context and predicts the most likely next word.

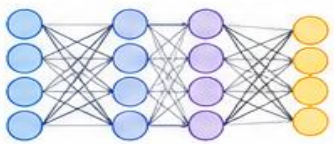
EVOLUTION OF LANGUAGE MODELS

1. Statistical Language Models



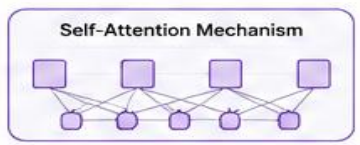
- Based on probability & statistics
- Uses n-grams, Markov models
- Limited context understanding

2. Neural Language Models



- Uses neural networks (RNN, LSTM, GRU)
- Captures longer dependencies
- Better understanding of context

3. Transformer-based Language Models



- Uses Transformer architecture
- Self-attention captures global context
- Highly parallel, efficient & scalable

This evolution ultimately led to the development of **Large Language Models (LLMs)**

WHY LANGUAGE MODELS ARE NECESSARY

Understand & Generate Human Language
Enables machines to understand, interpret, and generate human-like text.

Powering Modern NLP Applications
Foundation for autocomplete, spell checking, translation, summarization, Q&A, and chatbots.

Handle Large Amounts of Text Data
Learns from vast text corpora to capture knowledge, patterns, and relationships.

Adaptable & Scalable
Can be fine-tuned for various tasks and domains, scaling from small models to large models (LLMs).

Enables Conversational AI & LLMs
Essential for building intelligent assistants like ChatGPT, Claude, Gemini, and other LLM-based systems.

LARGE LANGUAGE MODELS (LLMs)

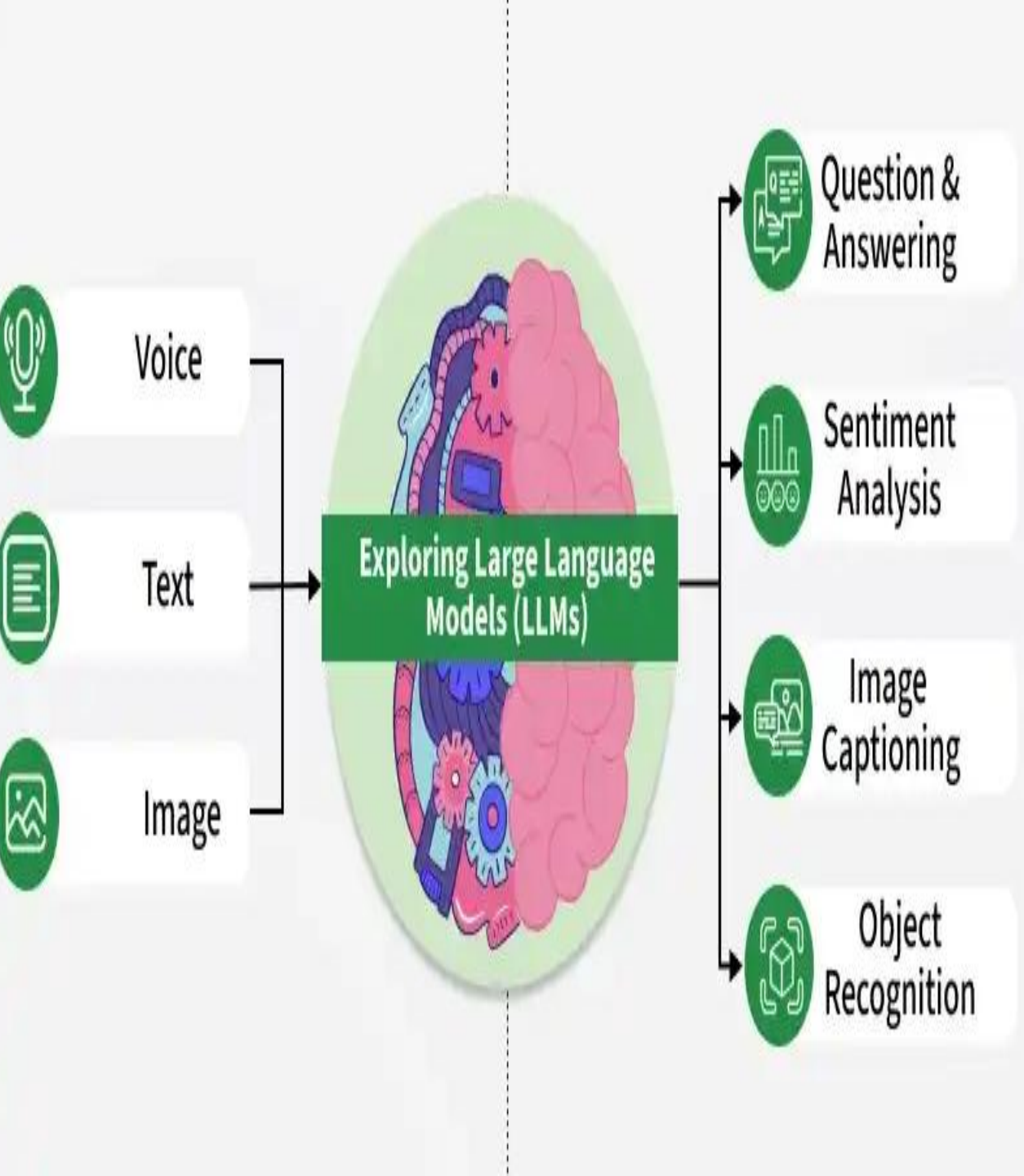


LLMs are Transformer-based models trained on massive amounts of data with billions of parameters. They can understand and generate human-like text across a wide range of tasks.

Examples: ChatGPT, Llama, Gemini, Claude, Mistral, Qwen, etc.

APPLICATIONS POWERED BY LANGUAGE MODELS





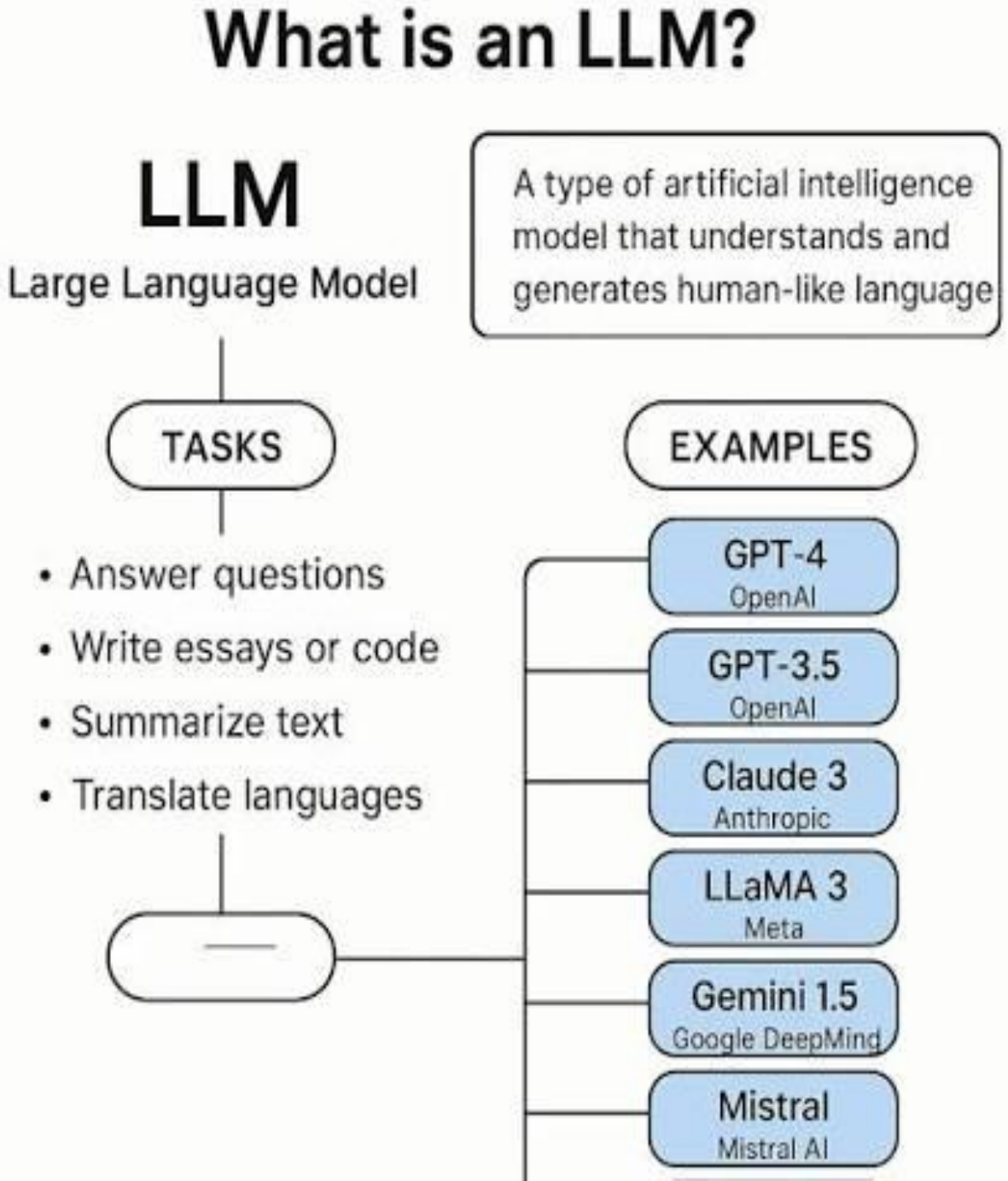
Large Language Model (LLM)

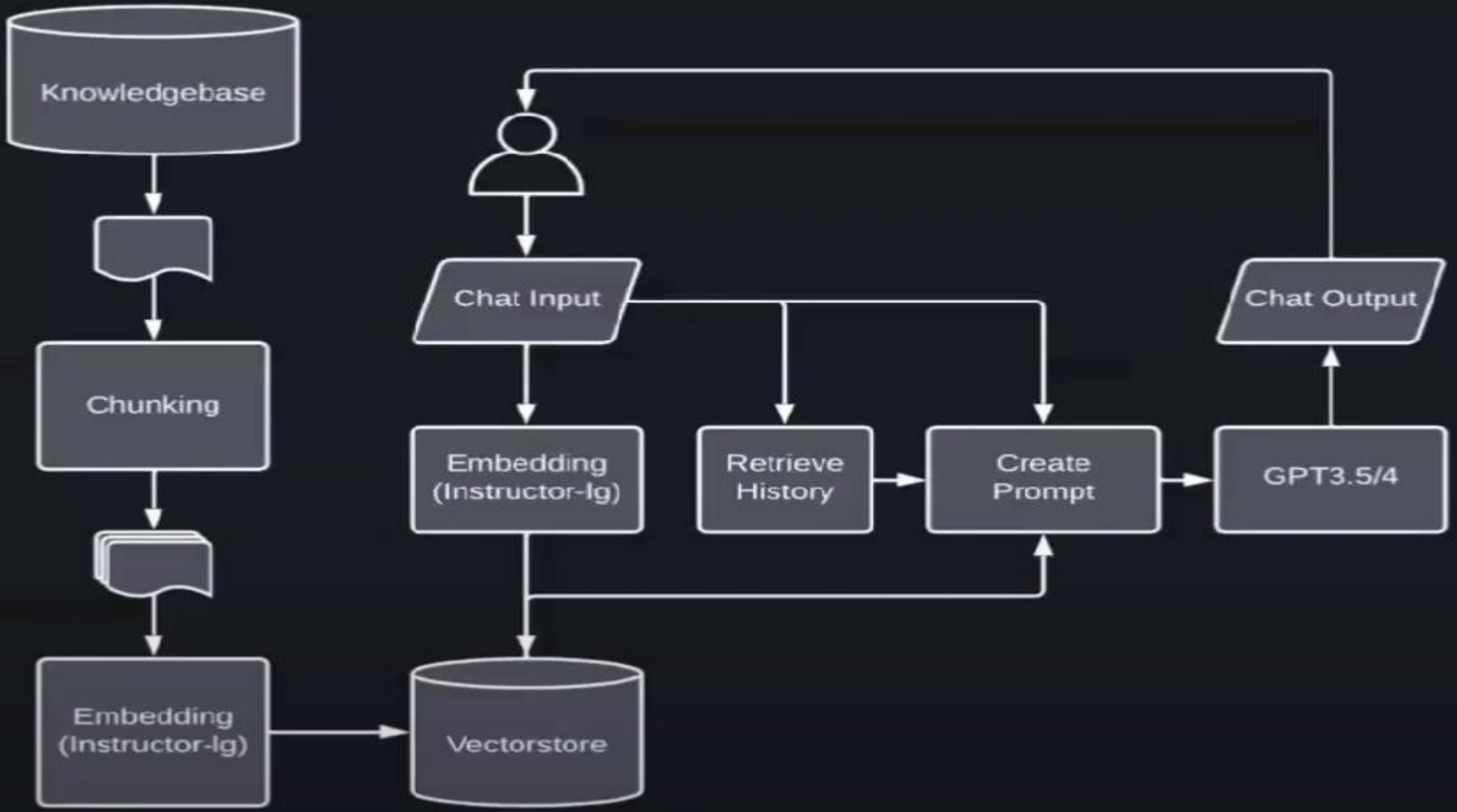
- Large Language Model (LLM) is an advanced type of Artificial Intelligence (AI) that is trained on massive amounts of text data to understand, generate, summarize, translate, and answer questions in human language.
- LLMs use Deep Learning (especially the Transformer architecture) to learn language patterns, grammar, context, and relationships between words.

- **Definition**
  - A Large Language Model (LLM) is an AI model trained on vast amounts of text data that can understand and generate human-like language.

How an LLM Works

- **Training on Large Datasets**
  - The model is trained using books, websites, a
- **Learning Language Patterns**
  - It learns grammar, vocabulary, sentence stru
- **Understanding the Prompt**
  - The user provides a prompt (question or inst
- **Predicting the Next Word**
  - The LLM predicts the most likely next word (response.
- **Generating the Response**
  - The model produces a coherent, human-like during training.




Knowledgebase
Chunking
Embedding
(Instructor-lg)
Vectorstore
Chat Input
Embedding
(Instructor-lg)
Retrieve
History
Create
Prompt
GPT3.5/4
Chat Output

Key Features of LLMs

- Understands natural language.
- Generates human-like text.
- Answers questions.
- Summarizes long documents.
- Translates between languages.
- Writes programming code.
- Generates emails, reports, and articles.
- Supports multi-turn conversations.
- Can explain complex topics in simple language.

LARGE LANGUAGE MODEL (LLM)

Understanding, Generating, and Empowering Human Language

A Large Language Model (LLM) is an AI model with billions to trillions of parameters, trained on massive amounts of text data to **understand** and **generate** human-like text.

WHAT IS AN LLM?

LLMs are advanced AI models based on deep learning (transformer architecture) trained on vast text data.

They learn the patterns, structure and context of human language and predict the next word (or token) in a sequence.

HOW DOES AN LLM WORK?

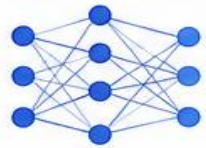
1. Input (Prompt)

User provides a prompt or question.

Example: "Explain photosynthesis in simple words."

→

2. Model Processing



LLM analyzes the context using billions of learned patterns.

→

3. Token Prediction

photosynthesis	0.21
is	0.16
process	0.35
the	0.12
...	

Predicts the next most likely token/word.

→

4. Output (Response)

Generates coherent, relevant and human-like text as the response.

The model repeats this process token by token to generate complete responses.

KEY CHARACTERISTICS OF LLMs

- **Trained on Massive Data** — Learn from huge amounts of text from books, websites, articles, code, and more.
- **Understand Context** — Capture long-range dependencies and understand the meaning and context of text.
- **Generate Human-like Text** — Produce fluent, coherent, and contextually relevant responses.
- **Adaptable & Versatile** — Can be fine-tuned or prompted for a wide variety of tasks and domains.
- **Scalable** — Performance improves with more data, compute and better architectures.
- **Zero/Low-shot Learning** — Can perform tasks with little or no task-specific training using instructions or examples.

HOW TO USE AN LLM (Typical Workflow)

1. **Define Task** — Identify what you want the model to do.
2. **Provide Prompt** — Give clear instructions or questions.
3. **Model Inference** — LLM processes the input and generates a response.
4. **Review & Refine** — Evaluate the output and refine the prompt if needed.
5. **Use Output** — Use the generated content in your application or workflow.

APPLICATIONS OF LLMs

- **Conversational AI** — Chatbots, virtual assistants, customer support.
- **Content Creation** — Blog posts, articles, marketing copy, stories, emails.
- **Language Translation** — Translate text between multiple languages accurately.
- **Summarization** — Summarize long documents, reports, and articles.
- **Information Retrieval** — Find and extract relevant information from large documents or knowledge bases.
- **Code Generation** — Generate, explain, debug and optimize code.
- **Education & Tutoring** — Explain concepts, answer questions, create quizzes.
- **Research & Analysis** — Analyze data, extract insights and assist in research workflows.

EXAMPLES OF POPULAR LLMs

GPT-4 / GPT-4o (OpenAI) · Gemini (Google) · Llama 3 (Meta) · Claude 3 (Anthropic) · Mistral Large (Mistral AI)

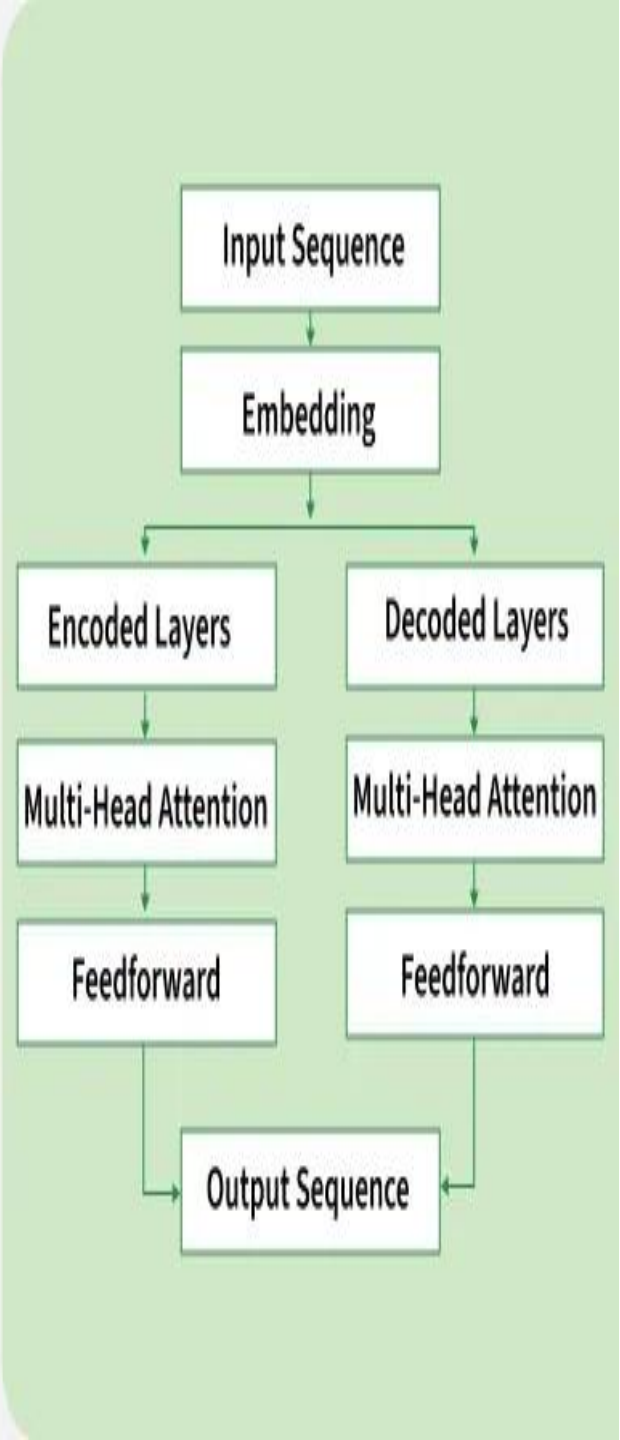
USE RESPONSIBLY

Always fact-check important information, ensure privacy and use AI ethically.

LLMs are transforming the way we interact with technology and information—enabling machines to understand, communicate, and create human-like text for countless real-world applications.

What are Transformers?

- Neural network architecture for NLP & vision
- Introduced in 2017: "Attention is All You Need"
- Processes whole sentences in parallel



Popular LLMs

- ChatGPT (OpenAI)
- Google Gemini
- Claude (Anthropic)
- Llama (Meta)
- Mistral
- DeepSeek
- Cohere Command

Transformers

